Artificial Intelligence

REGRESSION

Linear Regression • Gradient Descent • Logistic Regression

Classification and Regression

How much?

Input: advertising expenditure

Target: sales revenue

Which category?

Inputs: humidity and pressure

Target: Rain or No Rain

Advertising → **Sales**

Weather features → **Rain / No Rain**

Discuss: What values are allowed in each target space?

Supervised Learning

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$$

The calligraphic $\mathcal{D}$ helps distinguish the dataset from ordinary variables.

Training

Features + known targets

→ Learn model parameters

Prediction

New features + learned model

→ Estimate an unknown target

Q02 • QUICK CHECK

What is supplied to a trained model for a new prediction?

A. The correct target

B. The eventual prediction error

C. The new feature values

D. Newly fitted parameters for that one case

Classification or Regression?

Inputs	Target output	Your decision
Advertising spending	Sales revenue	?
Humidity + pressure	Rain / No Rain	?
Flower measurements	Iris species	?
Room location	Temperature	?

Q01 • QUICK CHECK

Rain is encoded as 1 and No Rain as 0. What is the task?

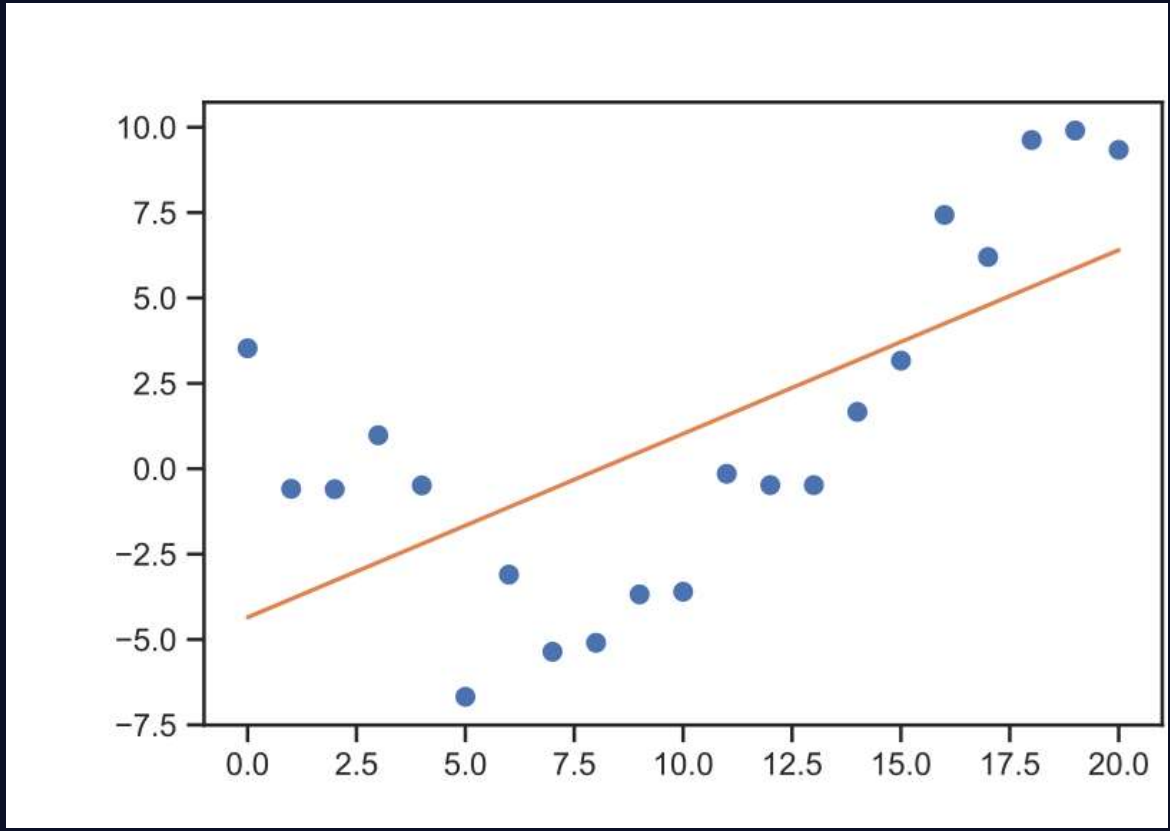
A. Regression: the targets are numbers

B. Classification: the numbers name categories

C. Clustering: there are two groups

D. It depends on the input type

Linear Regression



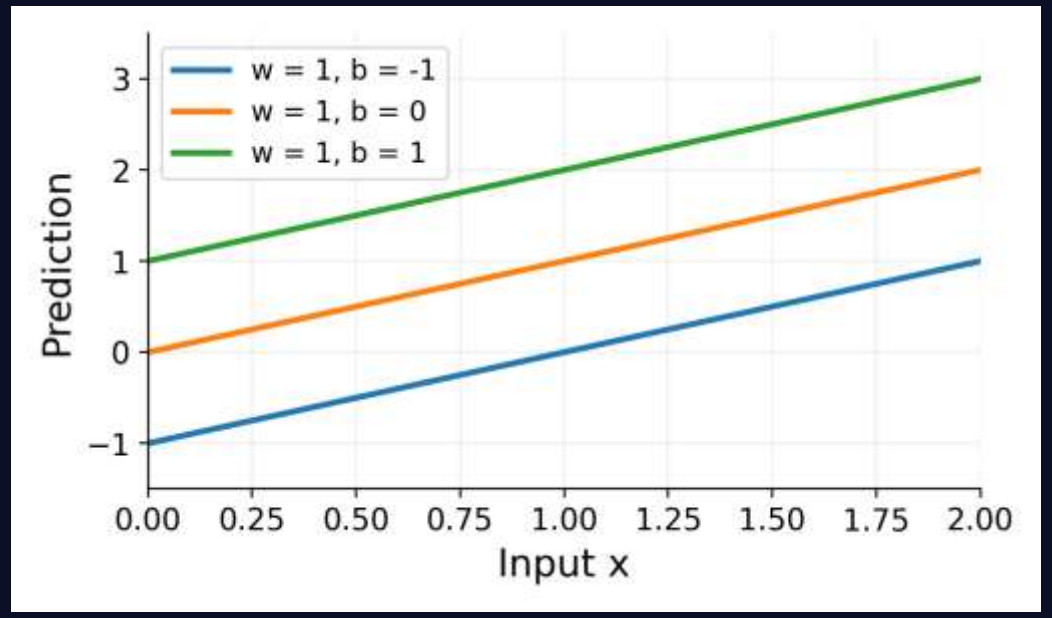
$$\hat{y} = wx + b$$

- **x** input feature
- **ŷ** model prediction
- **w** weight / slope
- **b** intercept / bias

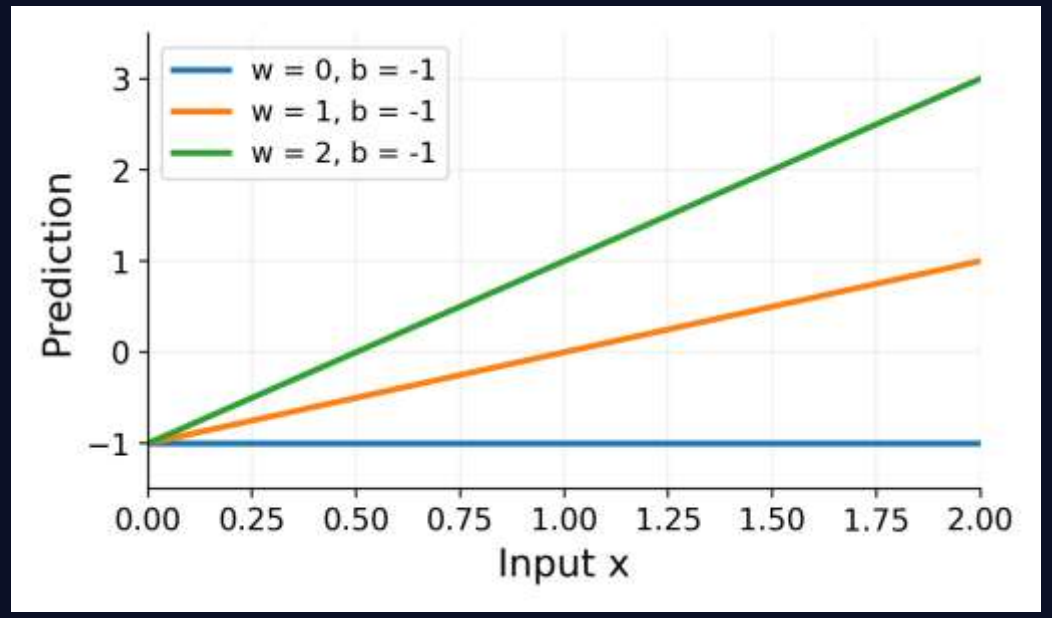
The model produces a prediction. Learning chooses w and b.

Slope and Intercept

HOLD W FIXED; CHANGE B



HOLD B FIXED; CHANGE W



Q03 • QUICK CHECK

Increase b while holding w fixed. What changes?

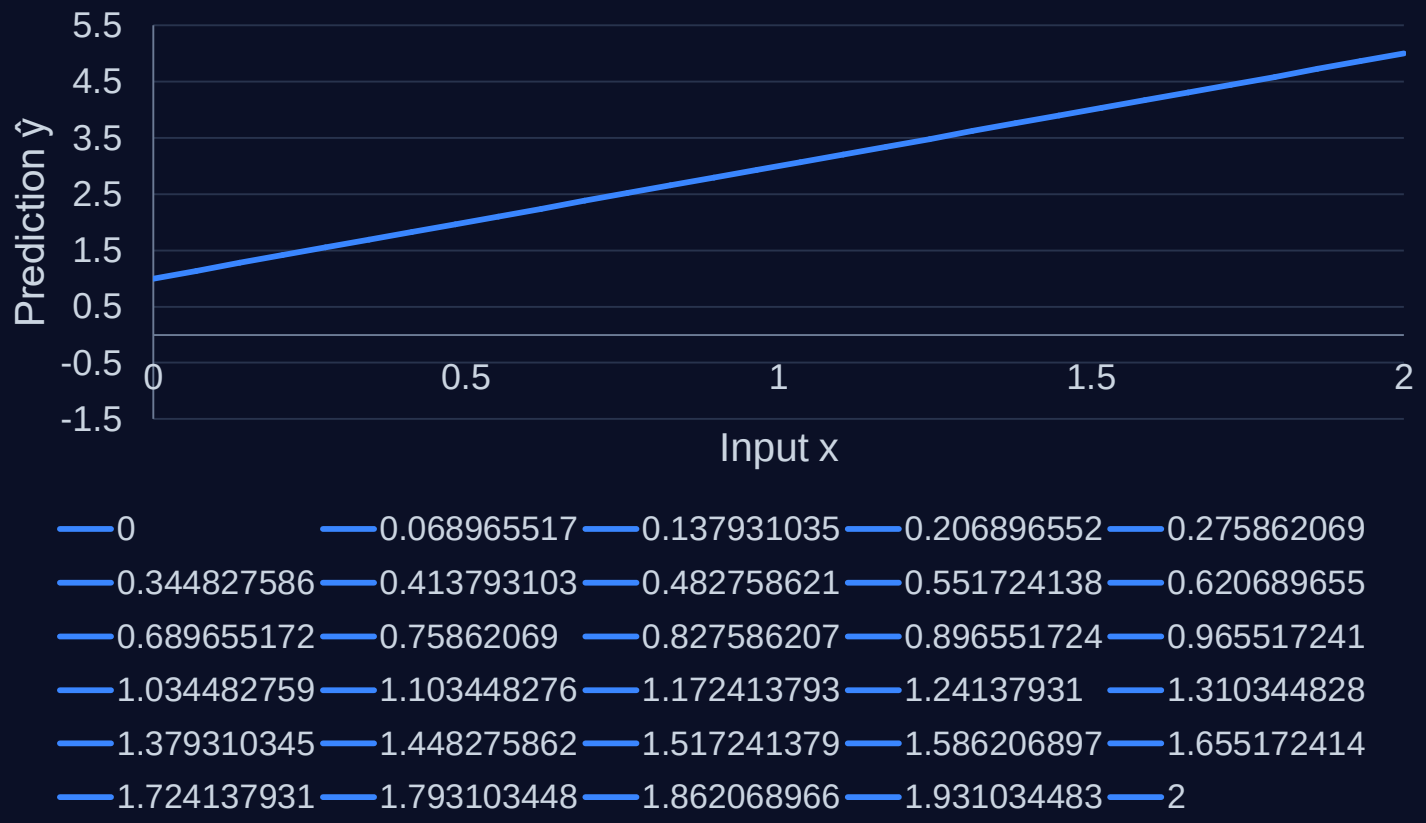
A. The line moves upward

B. The line becomes steeper

C. The number of features increases

D. The task becomes classification

Animation: Slope and Intercept



$$\hat{y} = wx + b$$

w = 2
b = 1

Hold the intercept at 1. The line pivots around its value at x = 0.

Step 6 of 6 · Click for next slide

Changing b shifts the line. Changing w changes its slope.

Prediction and Training

Predict with fixed parameters

Choose an input, then evaluate the function.

Learn from labeled examples

Compare predictions with targets, then improve the parameters.

$$w = 2, \quad b = -1, \quad x = 1$$

$$\hat{y} = 2(1) - 1 = 1$$

Training pairs (x, y)

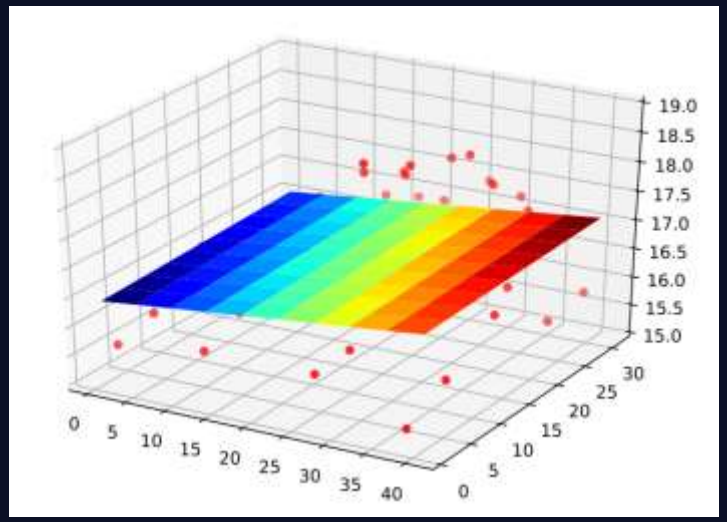
A choice of parameters (w, b)

Predictions and errors

Predicting evaluates a function. Training adjusts its parameters.

Multiple Linear Regression

$$\hat{y} = b + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d$$



Example: room temperature

Inputs: two location coordinates

Output: one temperature

A line becomes a plane.

Q04 • QUICK CHECK

Four input features and one intercept: how many parameters?

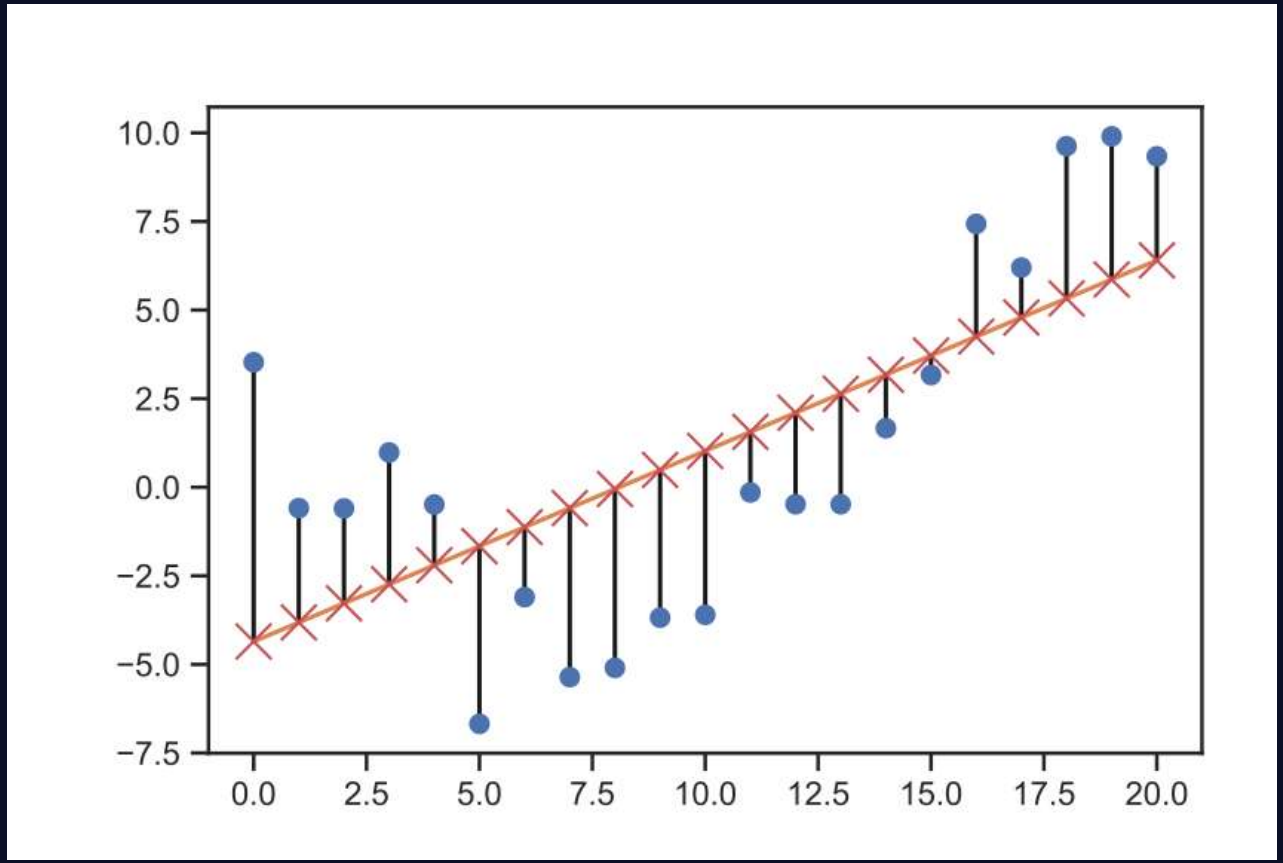
A. 4

B. 6

C. One per training observation

D. 5

Residuals



$$r_i \equiv y_i - \hat{y}_i$$

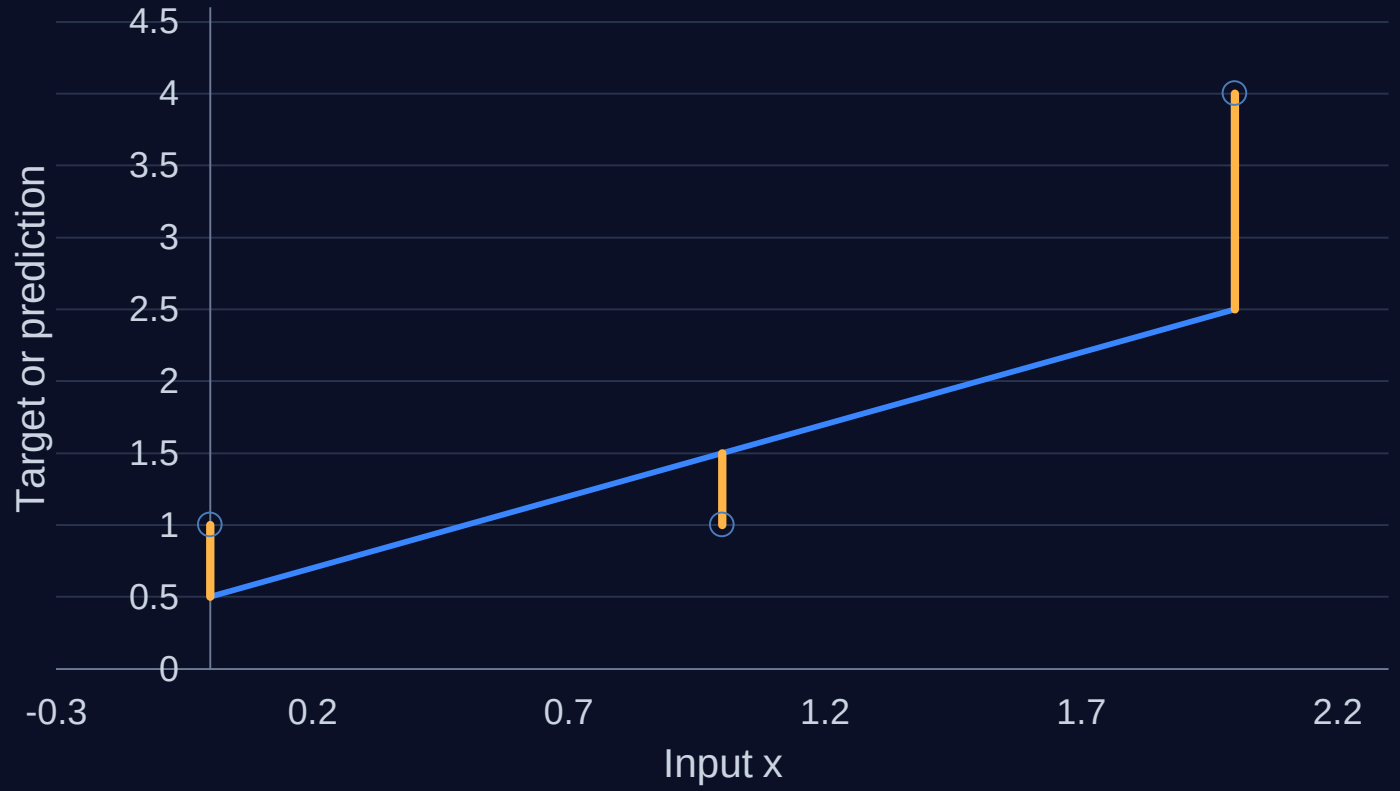
Observed target − prediction

Positive residual: prediction is too low.

Negative residual: prediction is too high.

Compare actual and predicted output at the same input x.

Animation: Residuals and Squared Error



$\hat{y} = x + 0.5$

Residual $r = y - \hat{y}$

$r1 = 0.5 \rightarrow r1^2 = 0.25$
$r2 = -0.5 \rightarrow r2^2 = 0.25$
$r3 = 1.5 \rightarrow r3^2 = 2.25$

MSE = 2.75 / 3 ≈ 0.917

Step 4 of 4 · Click for next slide

Each residual compares y and ŷ at the same input x.

Why We Square Errors

Add the signed residuals

One error can hide another.

Square before adding

Both errors now contribute.

$$(+1) + (-1) = 0$$

$$(+1)^2 + (-1)^2 = 2$$

Q05 • QUICK CHECK

Two residuals are +1 and −1. What is their MSE?

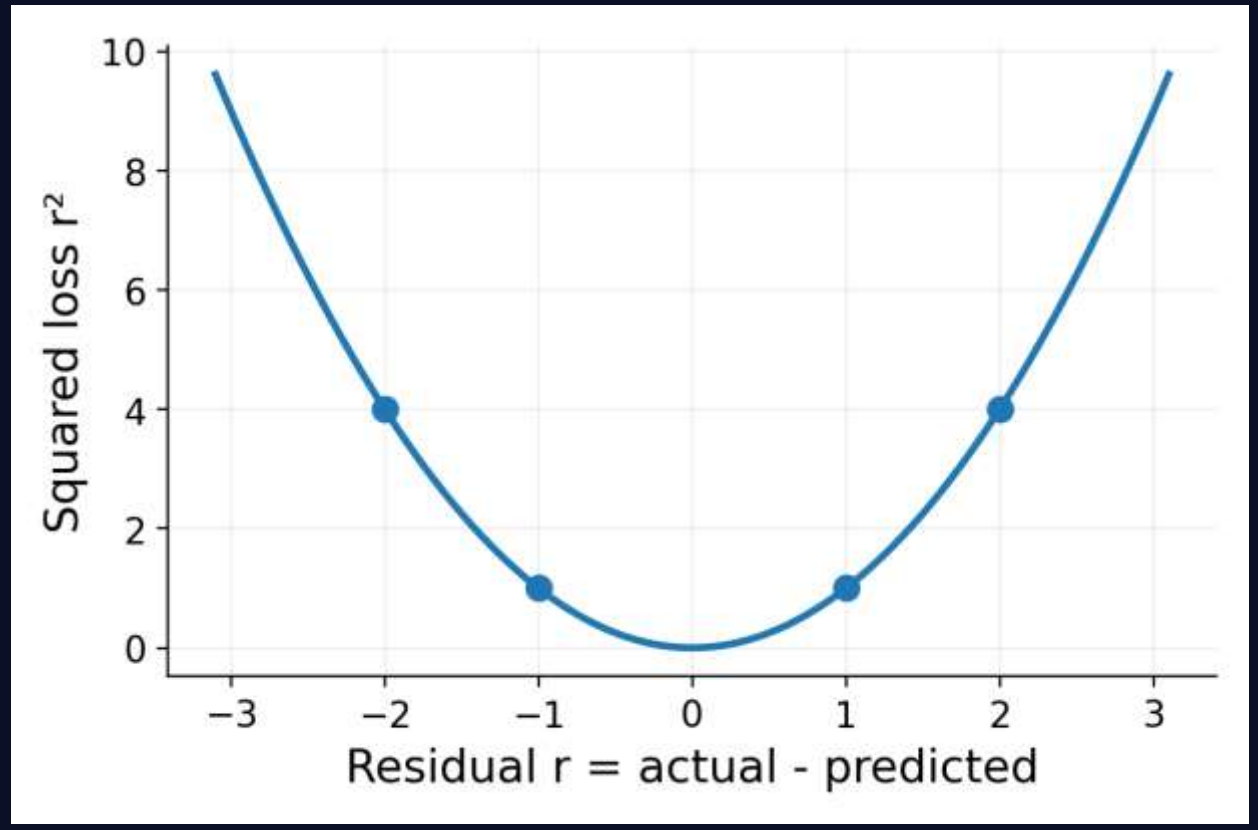
A. 0

B. −1

C. 1

D. 2

Squared Loss and Mean Squared Error



ONE EXAMPLE

$$\ell(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2$$

ALL N EXAMPLES

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

Double the error magnitude → four times the squared loss.

The Training Objective

$$J(w, b) = \frac{1}{2N} \sum_{i=1}^{N} (wx_i + b - y_i)^2$$

Fixed during an update

N and every observed pair (x_i, y_i)

Adjusted by learning

The weight w and intercept b

$J = \frac{1}{2}\,\mathrm{MSE}$

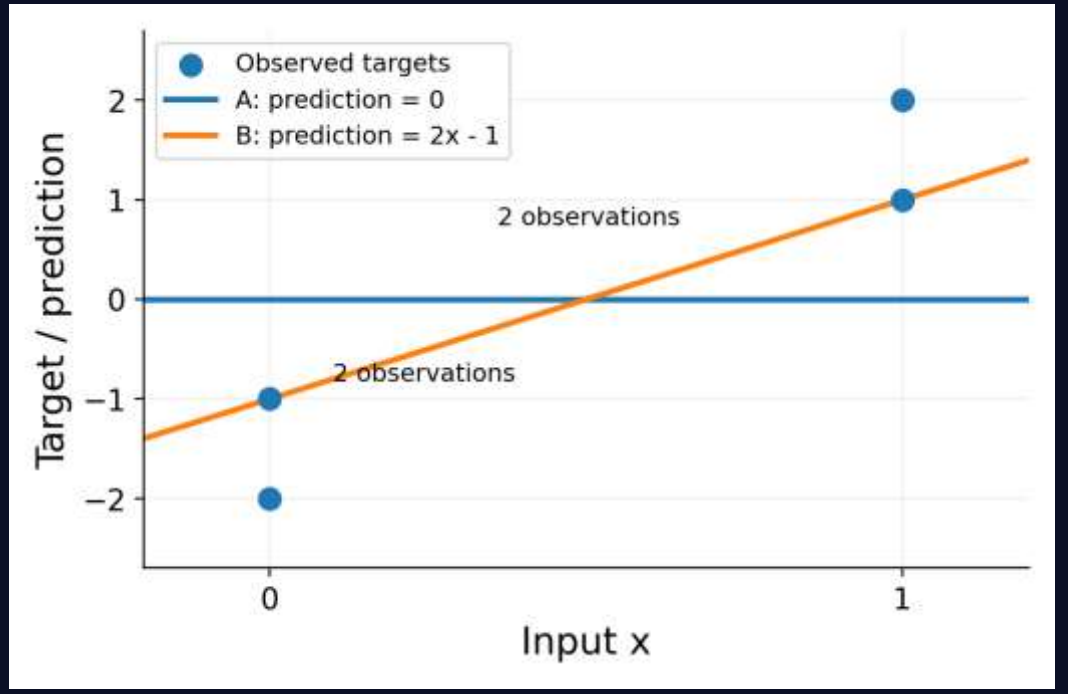
Same minimizer; a different gradient scale.

Comparing Candidate Lines

i	1	2	3	4	5	6
x_i	0	0	0	1	1	1
y_i	−1	−1	−2	1	1	2

KEEP ALL SIX OBSERVATIONS

$$A:\ \hat{y} = 0 \qquad B:\ \hat{y} = 2x - 1$$



Q06 • QUICK CHECK

A candidate has SSE = 2 over six observations. What is its MSE?

A. 0

B. 1/3

C. 1

D. 2

Learning as Optimization

$$(w^*, b^*) \in \operatorname{argmin}_{w,b} J(w, b)$$

Find the parameter values that give the smallest loss.

Minimize a loss or cost

Lower values are better.

Our choice: half-MSE.

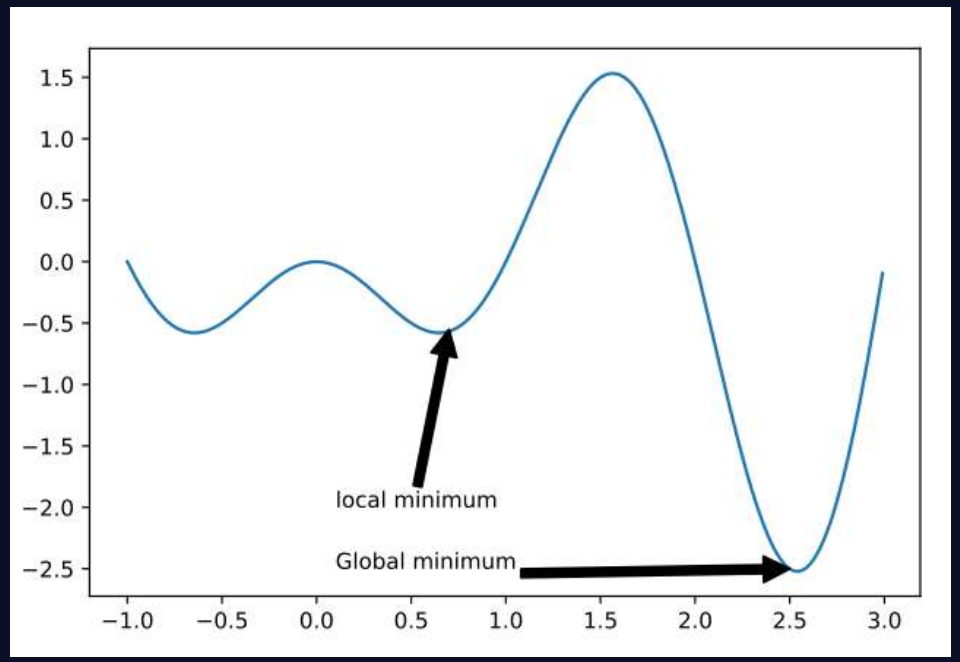
Maximize a score

Higher values are better.

Equivalent: minimize its negative.

Objective is the general term; a fitness score is another way to evaluate candidates.

Local and Global Minima



Local minimum

Best compared with nearby candidates.

Global minimum

Best compared with all allowed candidates.

Q07 • QUICK CHECK

A global minimum has the lowest objective value compared with…

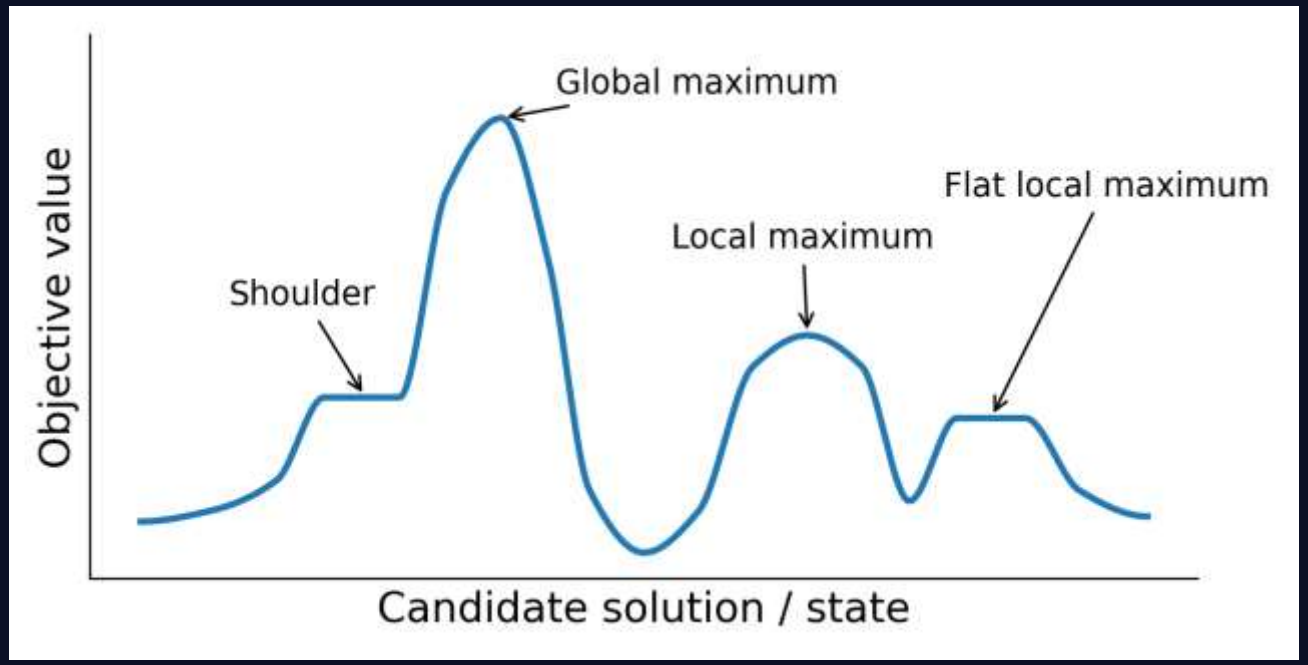
A. Only nearby candidates

B. Only the initial candidate

C. Every candidate in the allowed domain

D. Only candidates with objective zero

Local and Global Maxima



Q08 • QUICK CHECK

In a training-loss plot J(w) versus w, what is the horizontal axis?

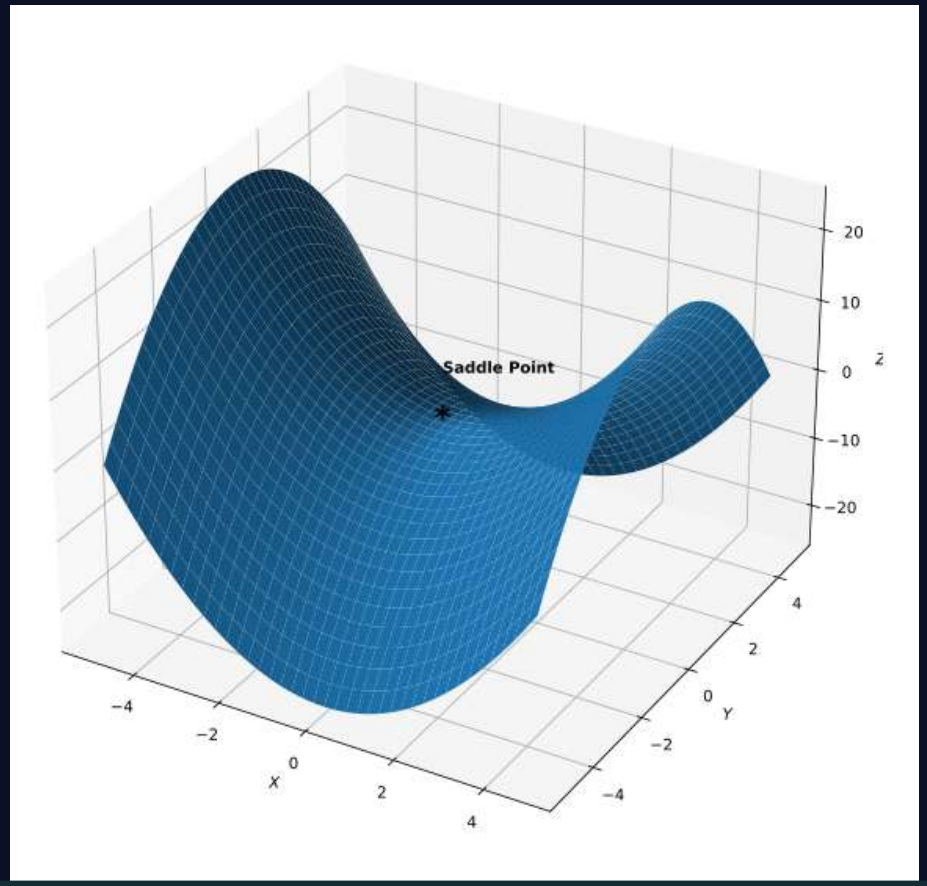
A. A model parameter

B. A training-example index

C. An observed target

D. Necessarily the feature x

Plateaus and Saddle Points



Plateau

A region where the objective changes little or not at all.

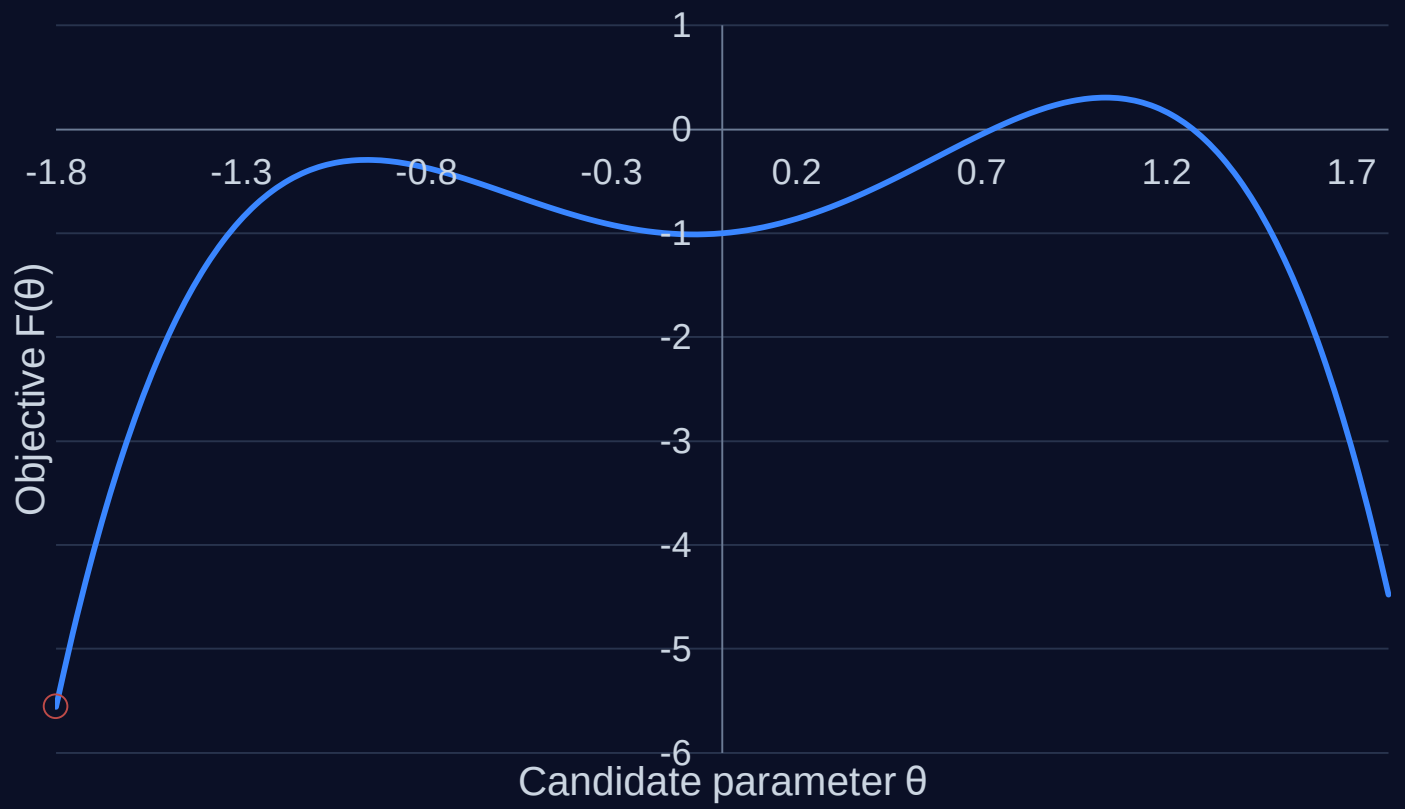
Saddle point

Uphill in some directions.

Downhill in other directions.

A zero gradient is not, by itself, proof of a minimum.

Animation: Local and Global Extrema



Global minimum

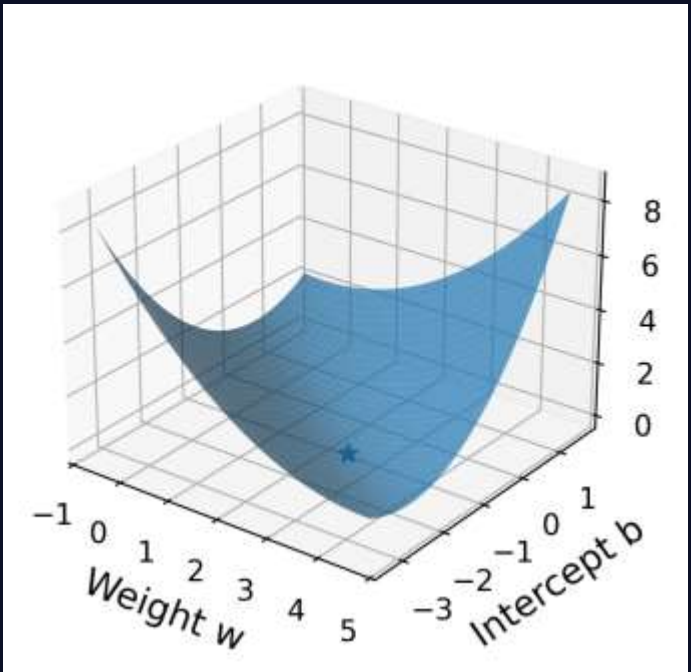
θ ≈ -1.80
F(θ) ≈ -5.56

The allowed domain is [−1.8, 1.8].
A global extremum can occur at a boundary.

Step 4 of 4 · Click for next slide

Local compares nearby candidates. Global compares the full allowed domain.

Convexity in Linear Regression



No worse local minima

Every local minimum is global.

Unique parameters need independent feature columns.

The minimum loss need not be zero.

Q09 • QUICK CHECK

For ordinary linear regression with squared-error loss…

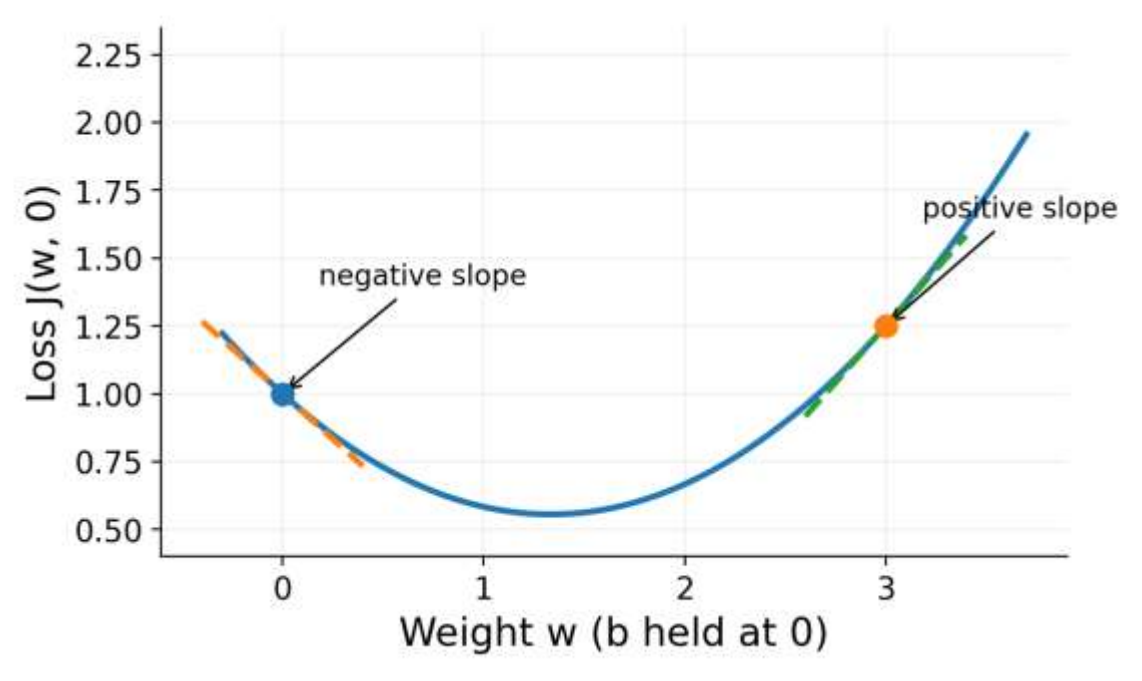
A. Bad isolated local minima are unavoidable

B. The minimum loss must be zero

C. Convexity alone guarantees unique weights

D. Every local minimum is global

The Derivative



$$\frac{\partial J}{\partial w}$$

Rate of change of loss as we change w

Hold the other parameter fixed.

Use a small local step.

Q10 • QUICK CHECK

If ∂J/∂w is positive, a sufficiently small descent step…

A. Increases w

B. Decreases w

C. Sets w to zero

D. Leaves w unchanged

Differentiation Rules

1 CONSTANTS

$$\frac{d}{d\theta}c = 0$$

2 SQUARE / POWER RULE

$$\frac{d}{du}u^2 = 2u$$

3 SUM AND CONSTANT FACTOR

$$\frac{d}{d\theta}[cf(\theta) + g(\theta)] = cf'(\theta) + g'(\theta)$$

4 CHAIN RULE

$$\frac{d}{d\theta}F(u(\theta)) = \frac{dF}{du}\frac{du}{d\theta}$$

Differentiate the outer function, then multiply by the derivative of the inner function.

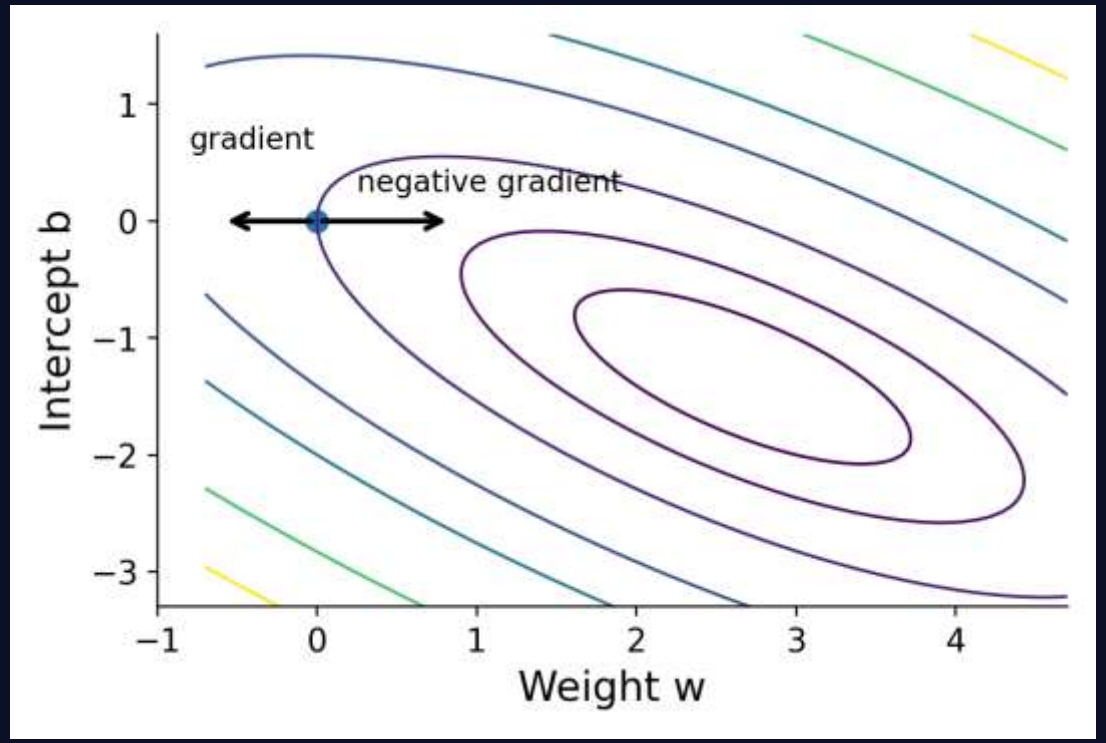
The Gradient

$$\nabla J(w, b) = \left(\frac{\partial J}{\partial w}, \frac{\partial J}{\partial b} \right)^T$$

Partial derivative

Change one parameter.

Hold the other fixed.



The gradient points uphill. Its negative points downhill locally.

Gradient Descent

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla J(\theta^{(t)})$$

Old parameters

Where we are now

Learning rate α > 0

How large a step to take

Current gradient

Which way is uphill

Q12 • QUICK CHECK

For a simultaneous update of w and b, compute both gradients…

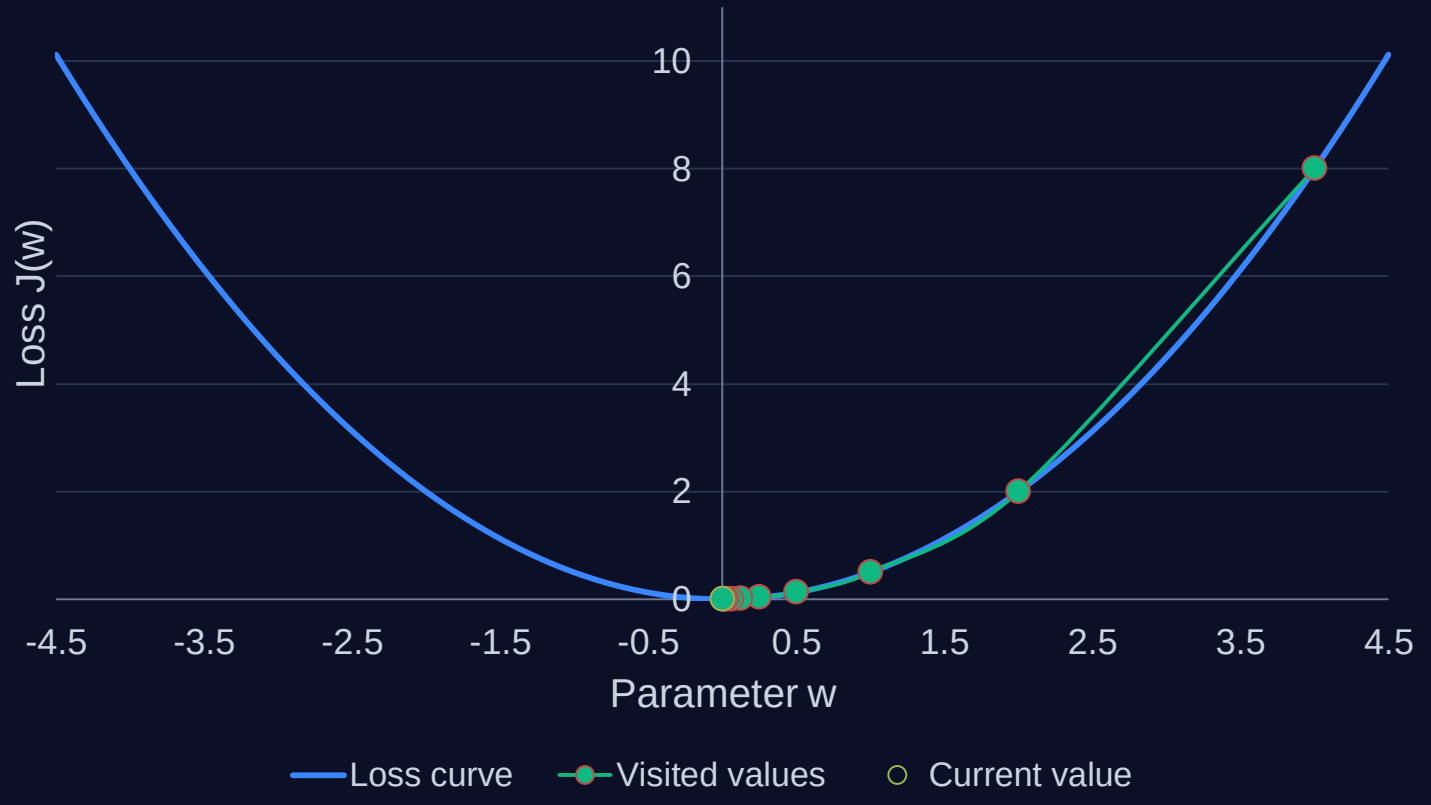
A. After updating w

B. After updating b

C. At different parameter states

D. At the same old parameter state

Animation: Gradient Descent



$J(w) = \frac{1}{2}w^2$

$dJ/dw = w$ $\alpha = 0.5$

Update 10
w = 0.0039
J = 0.000008

wnew = w − 0.5w
The steps shrink near 0.

Step 7 of 7 · Click for next slide

The negative gradient moves the parameter toward lower loss.

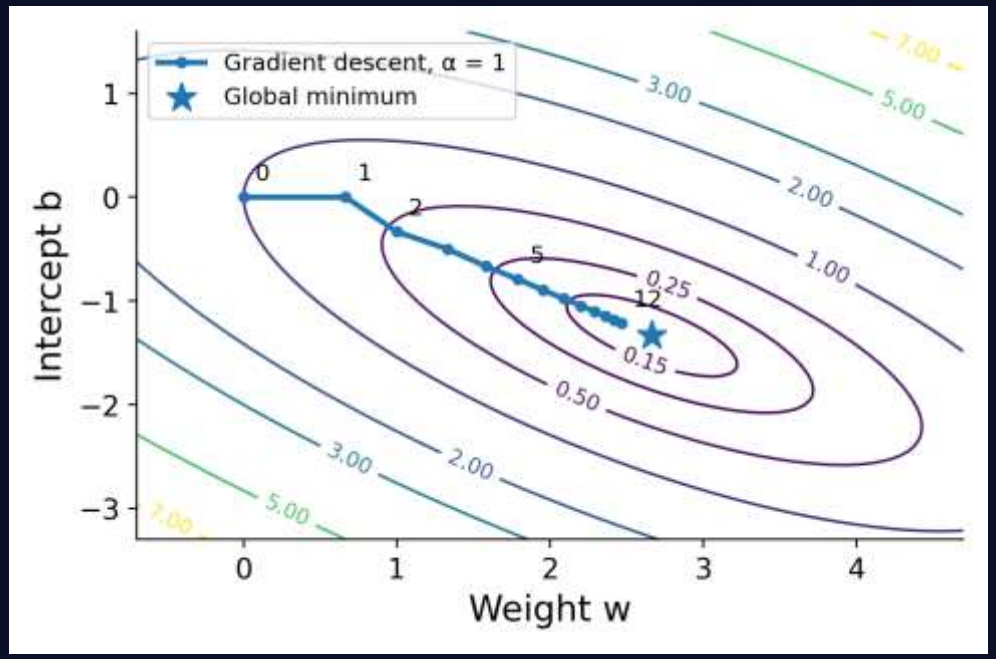
One Gradient Descent Update

1 PREDICT Use the current parameters.

2 MEASURE Calculate the loss and gradients.

3 UPDATE Replace the parameters simultaneously.

4 CHECK Recompute the loss; repeat.



Q11 • QUICK CHECK

w = 0, ∂J/∂w = −2/3, α = 1. What is the new w?

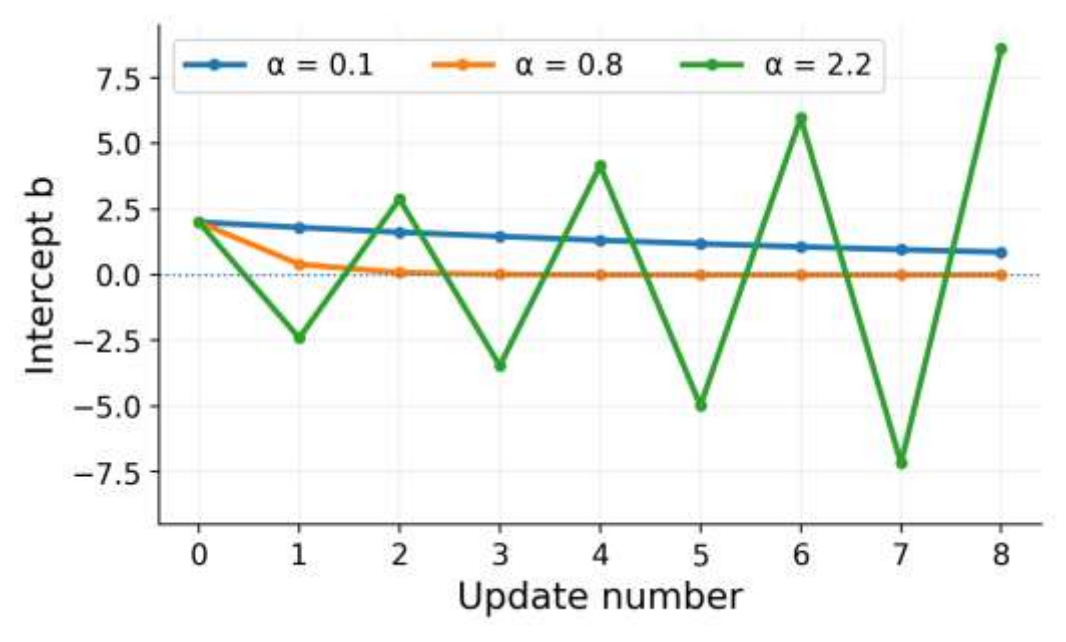
A. 2/3

B. −2/3

C. 0

D. 3/2

The Learning Rate



THIS ILLUSTRATION

Hold w = 0.

Start b = 2.

$$J(0, b) = 1 + \frac{b^2}{2}$$

$$b_{\text{new}} = (1 - \alpha)b$$

Q13 • QUICK CHECK

What can an excessively large learning rate cause?

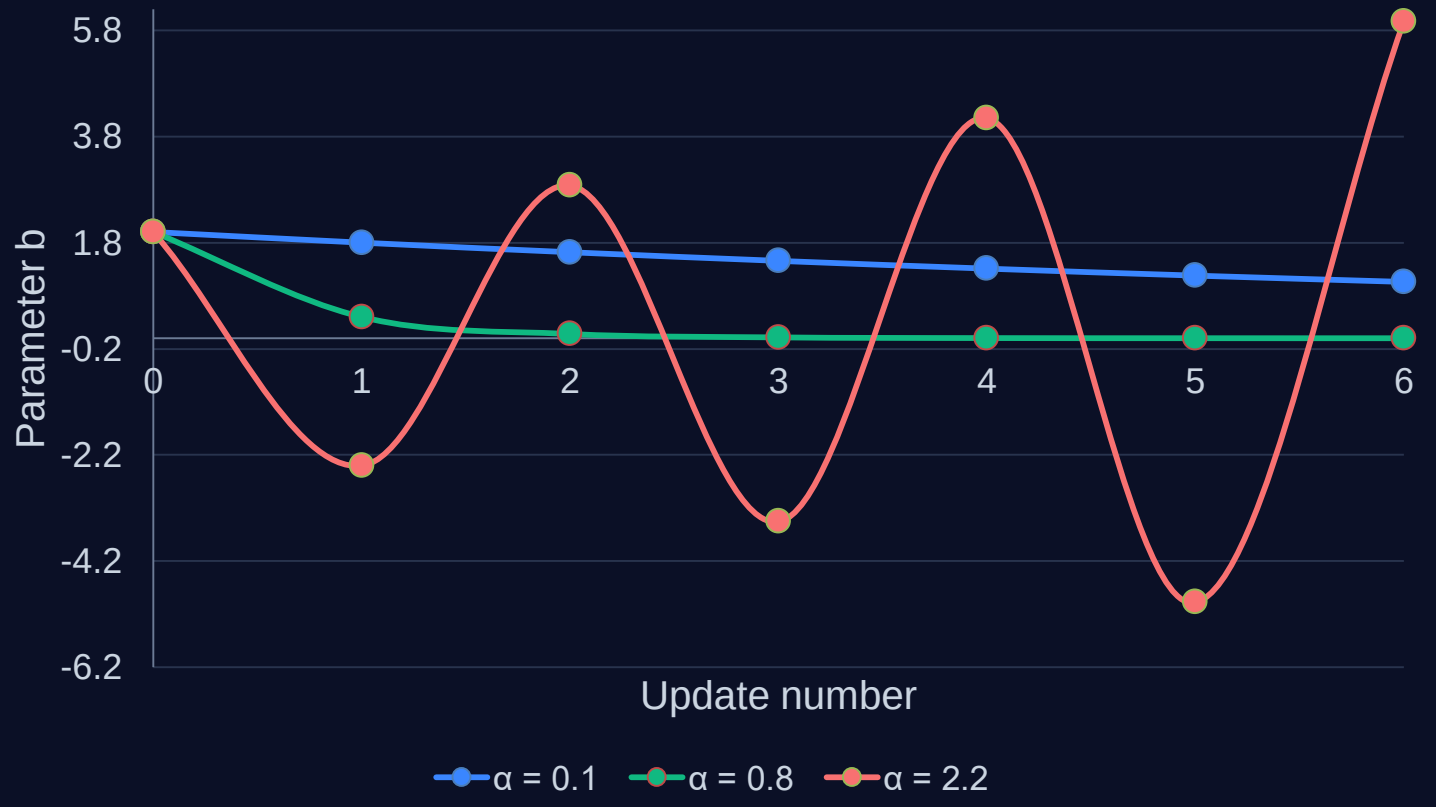
A. Guaranteed better predictions

B. Guaranteed faster convergence

C. Oscillation or divergence

D. Automatic changes to the target labels

Animation: Choosing a Learning Rate



Same start: b = 2

$J(0,b) = 1 + \frac{1}{2}b^2$

$b_{new} = (1 - \alpha)b$

α = 0.1 Slow progress

α = 0.8 Rapid convergence

α = 2.2 Growing oscillation

Step 7 of 7 · Click for next slide

Small steps converge slowly. An oversized learning rate can diverge.

Deriving the Gradients

$$e_i = wx_i + b - y_i = -r_i, \qquad L_i = \frac{1}{2}e_i^2, \qquad J = \frac{1}{N}\sum_i L_i$$

WITH RESPECT TO W

$$\frac{\partial L_i}{\partial w} = \frac{1}{2}(2e_i)\frac{\partial e_i}{\partial w}$$

$$\frac{\partial e_i}{\partial w} = x_i$$

$$\frac{\partial L_i}{\partial w} = e_i x_i$$

$$\frac{\partial J}{\partial w} = \frac{1}{N}\sum_{i=1}^{N} e_i x_i$$

WITH RESPECT TO B

$$\frac{\partial L_i}{\partial b} = \frac{1}{2}(2e_i)\frac{\partial e_i}{\partial b}$$

$$\frac{\partial e_i}{\partial b} = 1$$

$$\frac{\partial L_i}{\partial b} = e_i$$

$$\frac{\partial J}{\partial b} = \frac{1}{N}\sum_{i=1}^{N} e_i$$

A Full-Batch Update

$$w^{(0)} = 0, \quad b^{(0)} = 0, \quad \alpha = 1, \quad N = 6$$

i	x_i	y_i	$\hat{y}_i$	e_i	$e_i x_i$
1	0	−1	0	1	0
2	0	−1	0	1	0
3	0	−2	0	2	0
4	1	1	0	−1	−1
5	1	1	0	−1	−1
6	1	2	0	−2	−2

SUM, THEN AVERAGE

$$\sum_i e_i x_i = 0 + 0 + 0 - 1 - 1 - 2 = -4$$

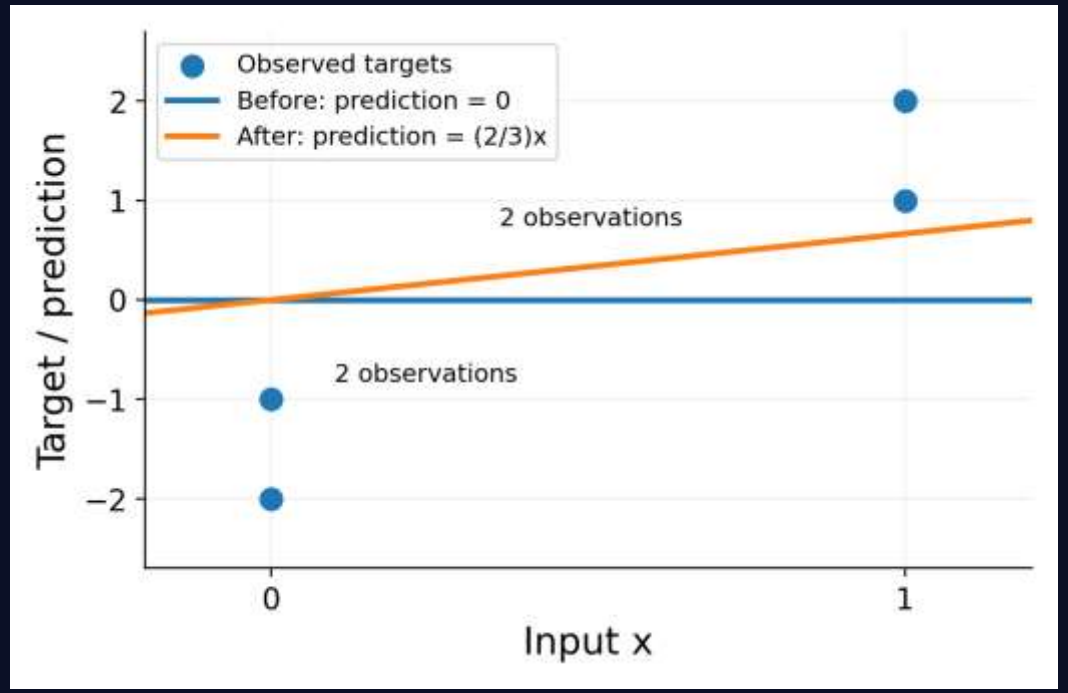
$$g_w = \frac{-4}{6} = -\frac{2}{3}$$

$$\sum_i e_i = 1 + 1 + 2 - 1 - 1 - 2 = 0$$

$$g_b = \frac{0}{6} = 0$$

$$w^{(1)} = 0 - 1(-2/3) = 2/3, \qquad b^{(1)} = 0 - 1(0) = 0$$

Checking the Updated Loss



RECOMPUTE WITH THE NEW PARAMETERS

$$\mathrm{SSE} = 1 + 1 + 4 + \frac{1}{9} + \frac{1}{9} + \frac{16}{9} = 8$$

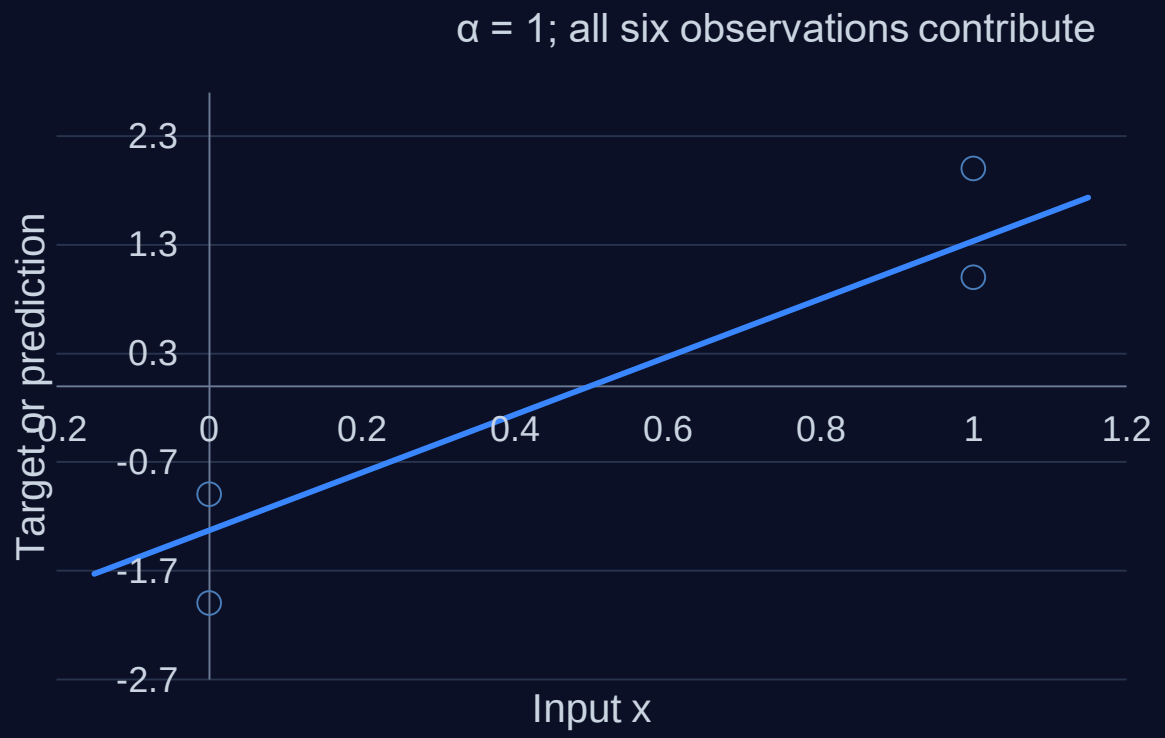
$$J_{\text{new}} = \frac{8}{2 \times 6} = \frac{2}{3} < 1$$

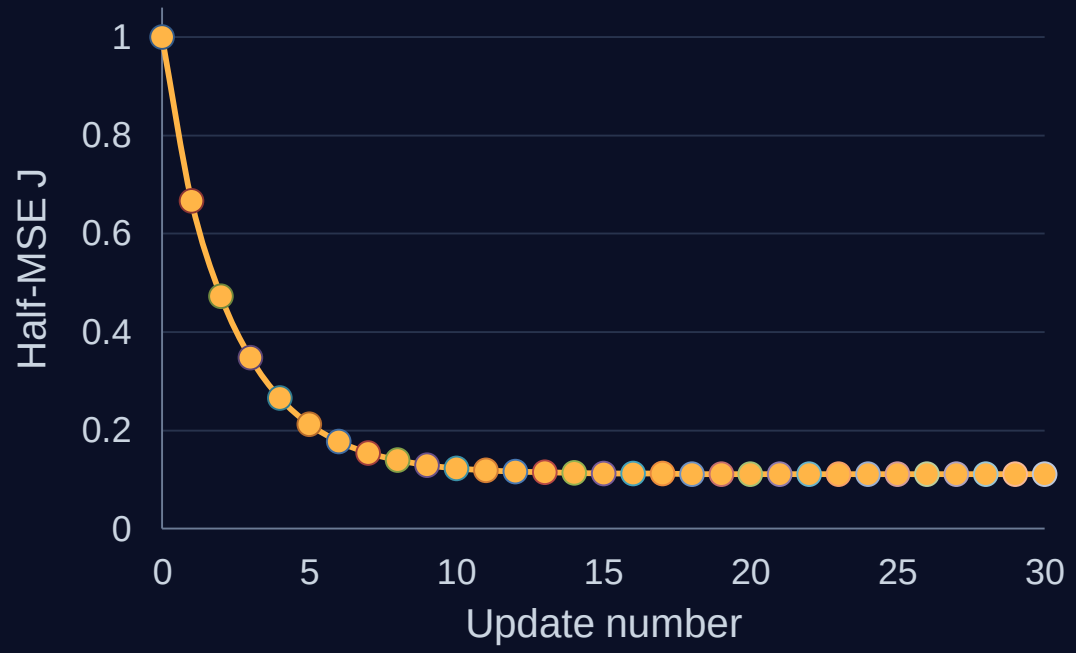
Better does not mean finished.

GROUP CHALLENGE • 4 MINUTES

Start from w = 2/3, b = 0, α = 1. Recompute both gradients, update, and calculate J.

Animation: Fitting the Line and Reducing Loss





Update 30 **w = 2.662** **b = -1.331** **J = 0.1111**

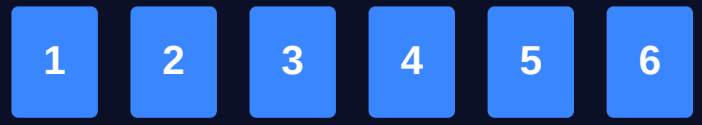
Step 7 of 7 · Click for next slide

Both gradients use the same old parameters. The best-fit loss is above zero.

Batch, Stochastic, and Mini-Batch

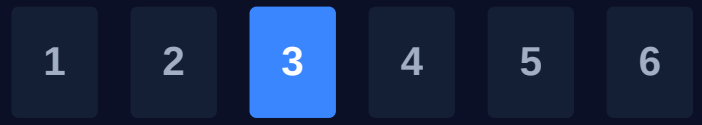
Full batch

All N examples



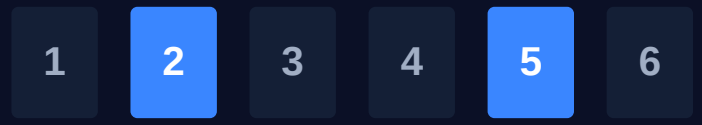
Stochastic

One sampled example



Mini-batch

A sampled subset



Training checks

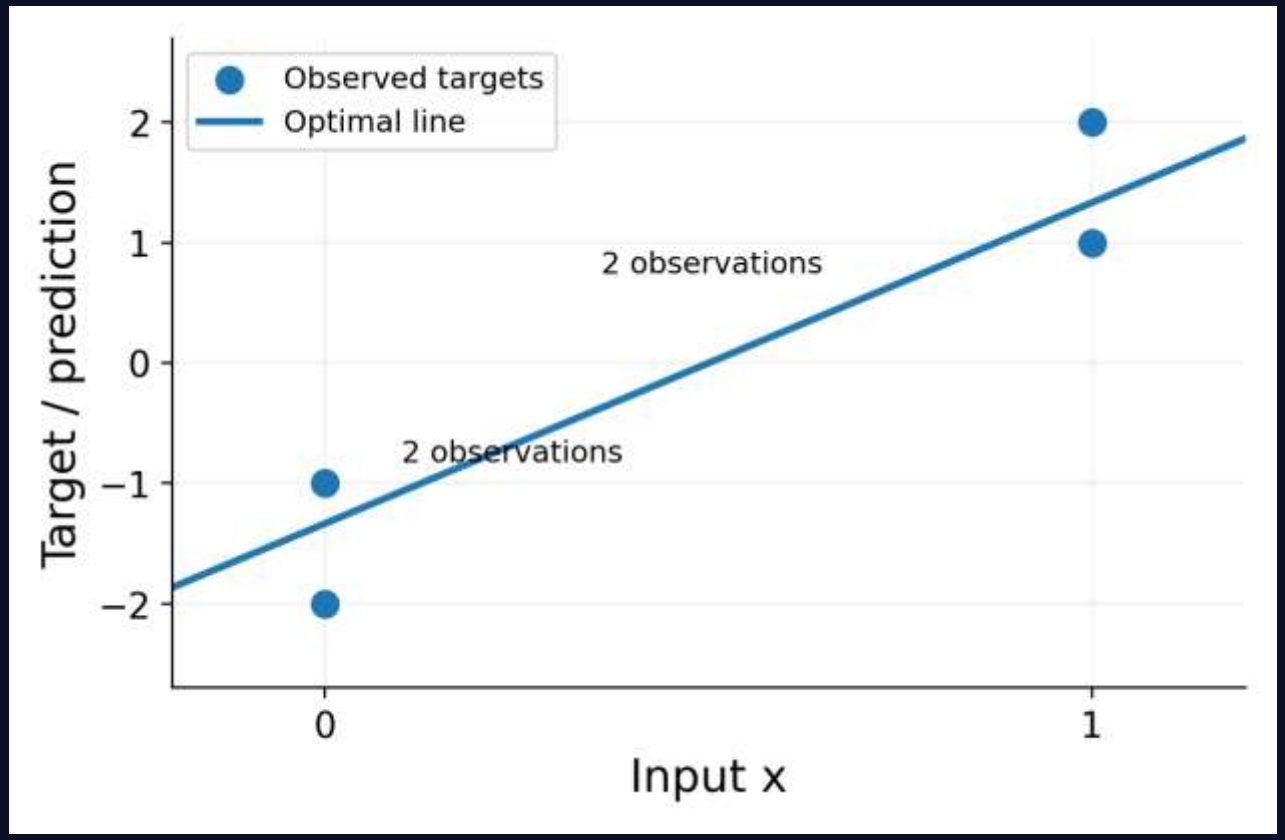
Inspect loss history.

Check the learning rate.

Standardize uneven feature scales.

Linear regression is a model. Gradient descent is one way to fit it.

Generalization



What the optimizer achieved

A low objective on the training examples.

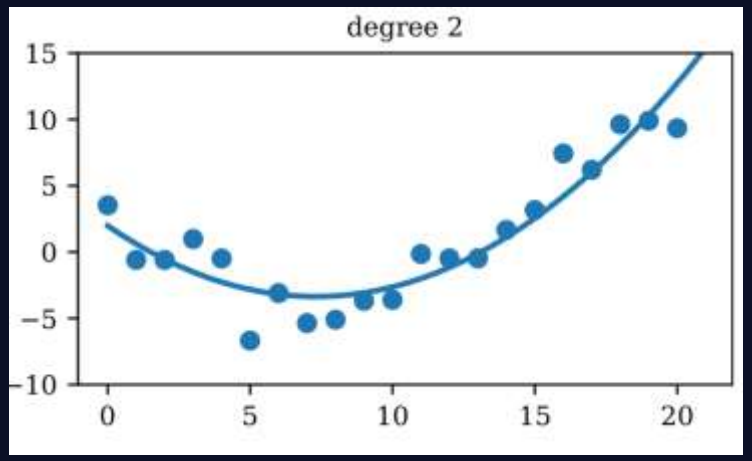
What still needs testing

Useful predictions on unseen examples.

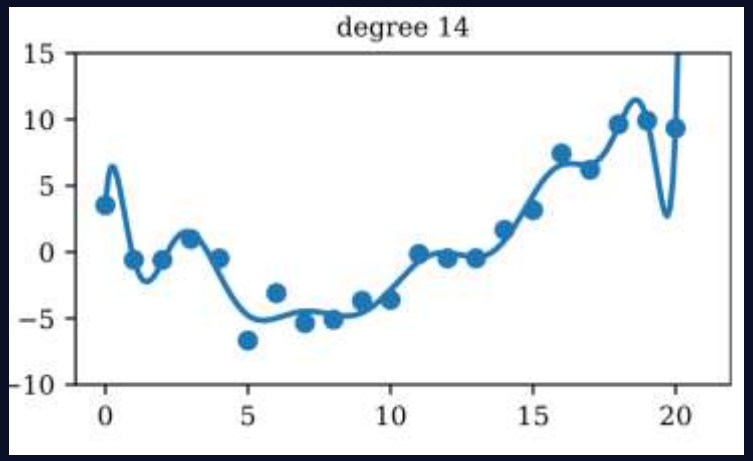
A minimum training loss is not a guarantee of generalization.

Model Flexibility and Overfitting

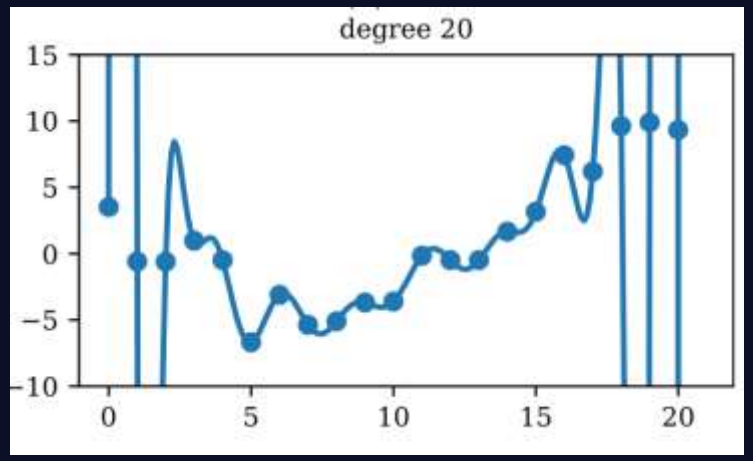
DEGREE 2



DEGREE 14



DEGREE 20



Same 21 observations. Increasing polynomial degree adds flexibility.

Vote: Which curve would you trust on new inputs? Explain your choice.

Training, Validation, and Test Sets

Training set

Fit w and b.

Learn from labeled examples.

Validation set

Choose model complexity and hyperparameters.

Test set

Estimate final performance on held-out examples.

Q14 • QUICK CHECK

Which workflow separates model selection from final evaluation?

A. Choose by training error only

B. Repeatedly tune on the test set

C. Use test labels during weight updates

D. Select on validation; evaluate on held-out test

Regularization

$$J_{\text{total}} = J_{\text{fit}} + \lambda\, C(\theta), \qquad \lambda \geq 0$$

Fit the observed data

Reward accurate predictions on the training examples.

Penalize complexity

Discourage overly flexible parameter settings.

Discuss: Is stronger regularization always better? Why not?

Binary Classification



Setosa | Versicolor | Virginica

BINARY TARGET

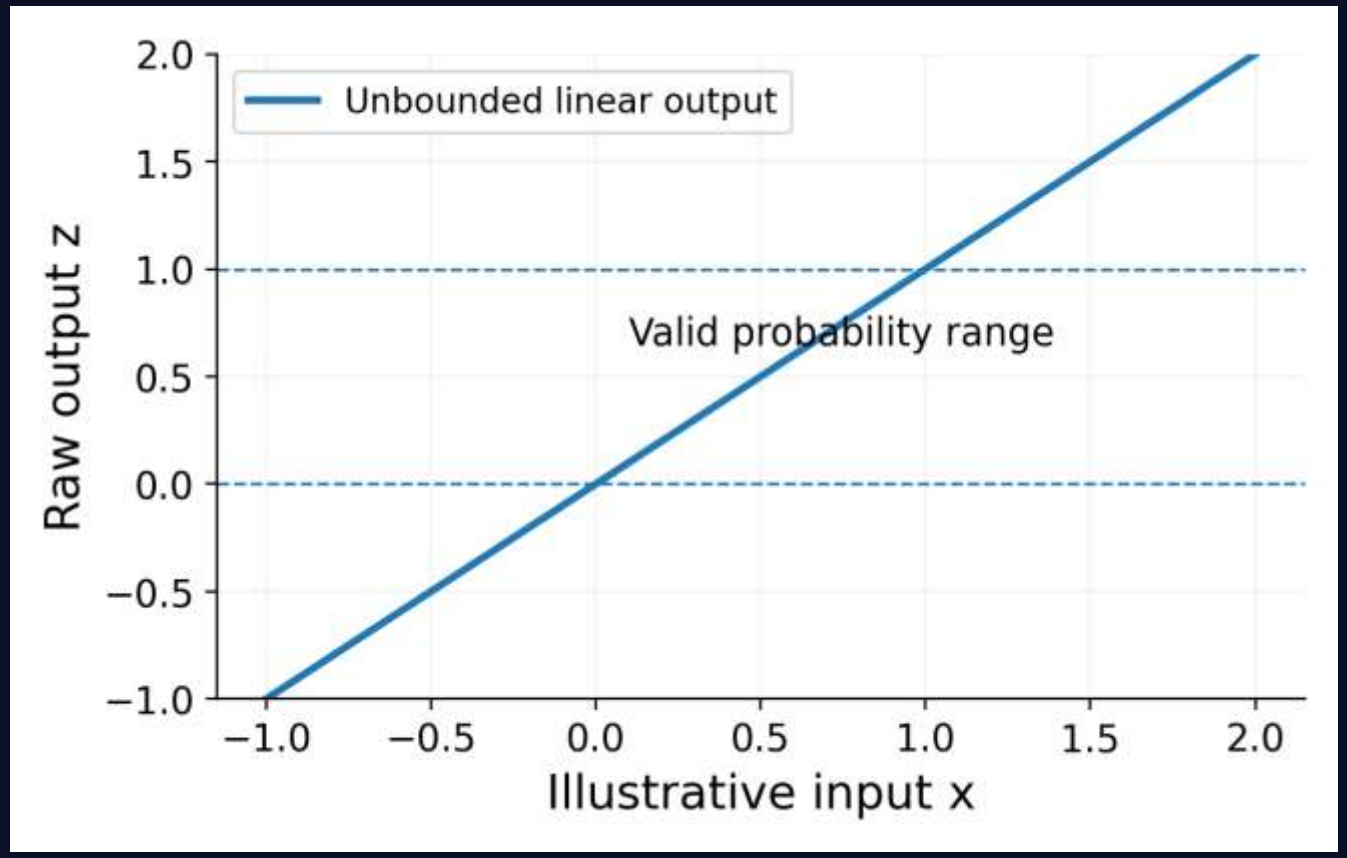
1 = Virginica

0 = Not Virginica

Input in our example: petal width (cm).

Target: a class label, not a flower measurement.

Linear Scores and Probabilities



LINEAR SCORE

$$z = b + \mathbf{w}^{\top}\mathbf{x}$$

REQUIRED FOR A PROBABILITY

$$0 \leq p \leq 1$$

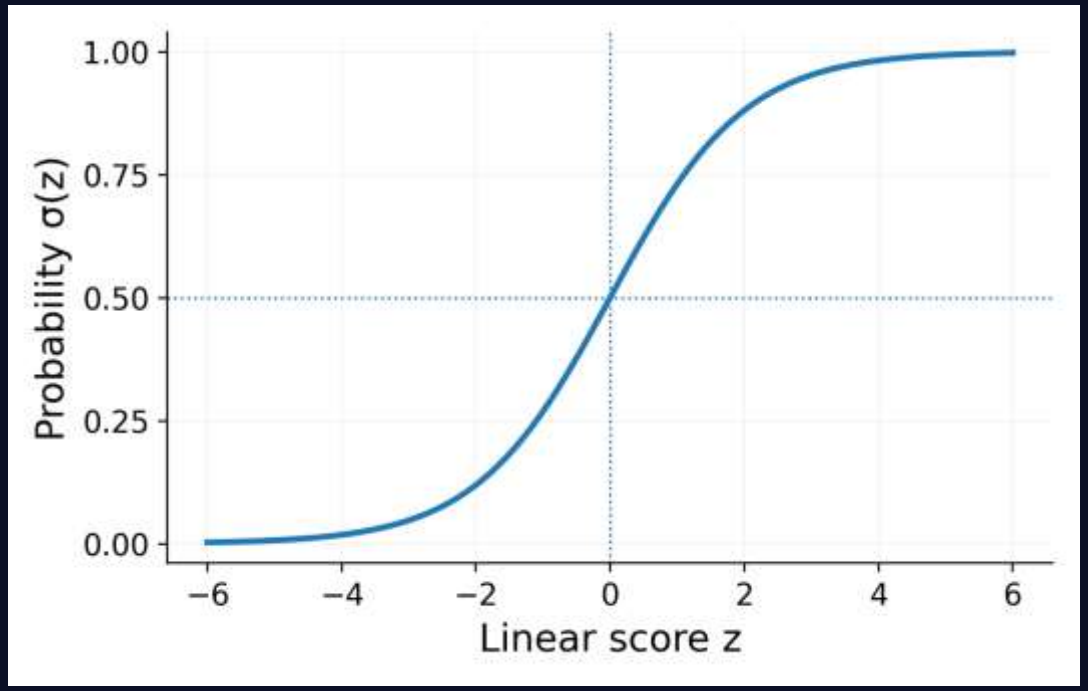
Keep the linear score. Transform it before calling it a probability.

The Sigmoid Function

$$z = b + \mathbf{w}^{\top}\mathbf{x}$$

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

p = estimated probability of class 1



Q15 • QUICK CHECK

What does the sigmoid output p represent in binary logistic regression?

A. A predicted numerical amount

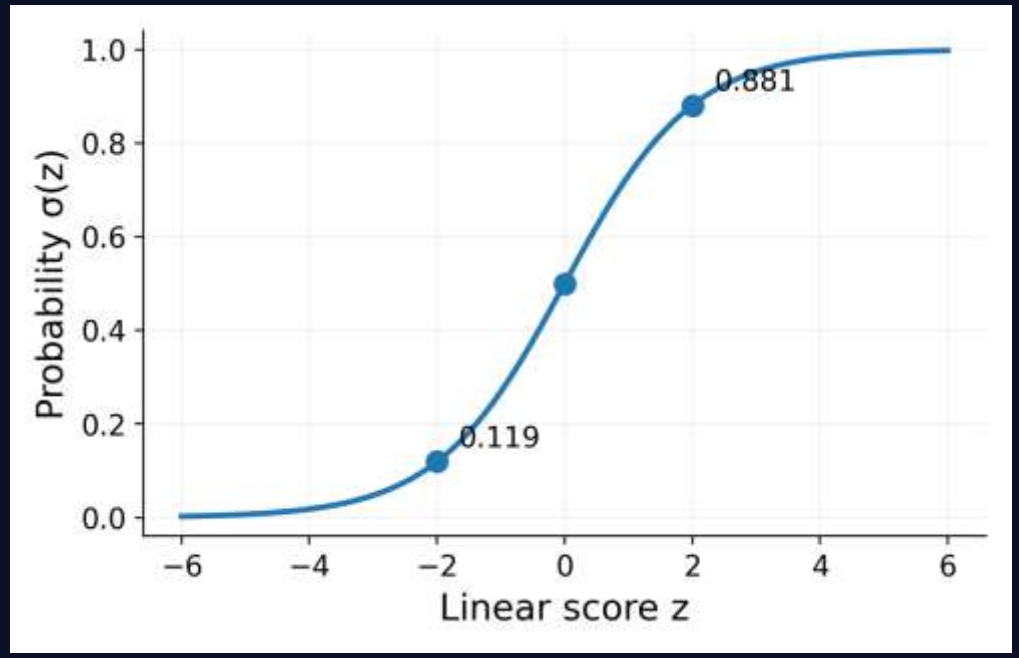
B. The estimated probability of class 1

C. The median input

D. The loss gradient

Evaluating the Sigmoid

Score z	Substitution	Probability
−2	$1 / (1 + e^{2})$	≈ 0.119
0	$1 / (1 + e^{0})$	?
2	$1 / (1 + e^{-2})$	≈ 0.881



Q16 • QUICK CHECK

When the linear score z = 0, what is σ(z)?

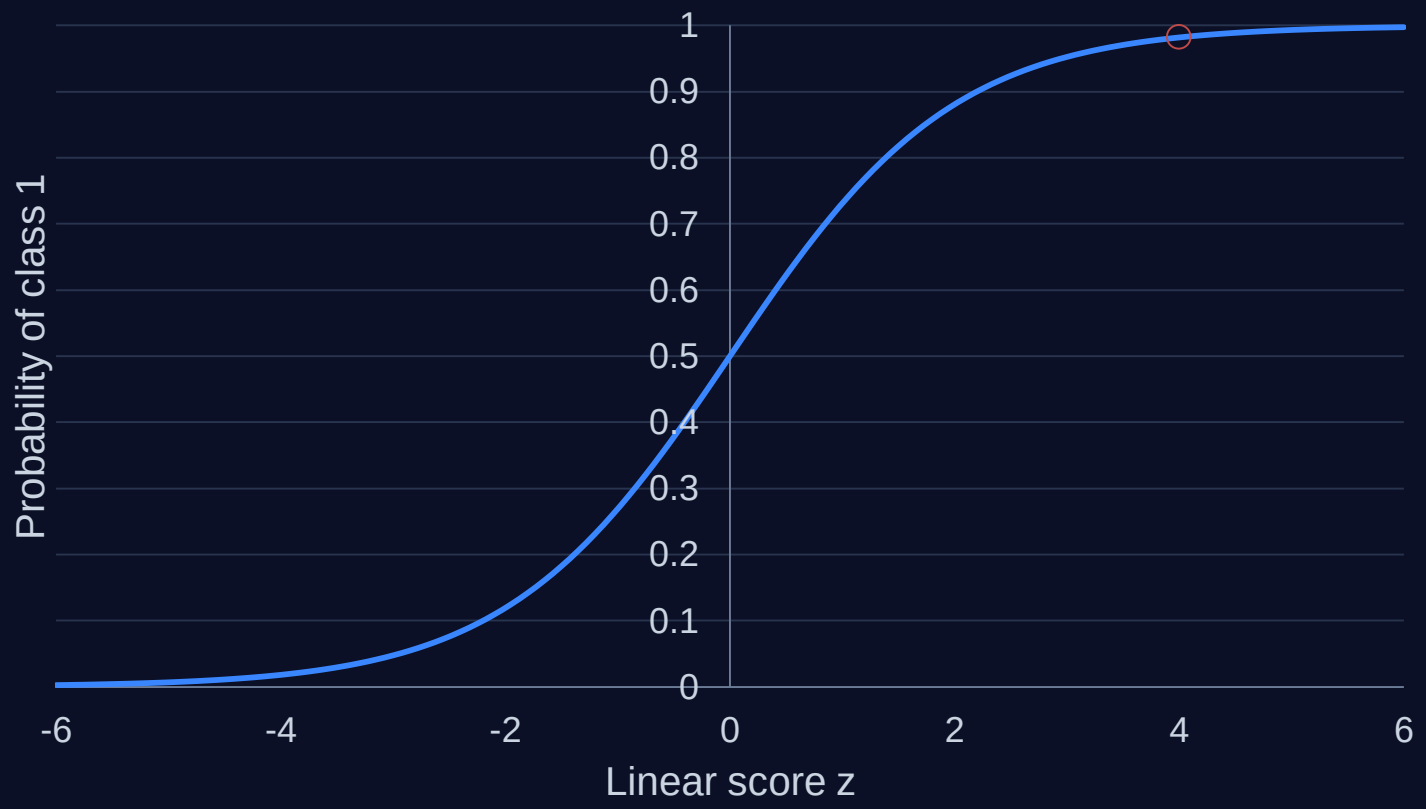
A. 0

B. 1

C. 0.5

D. Undefined

Animation: A Score Becomes a Probability



$p = 1 / (1 + e^{-z})$

z = 4

p ≈ 0.982

A positive score gives a probability above 0.5.

Step 5 of 5 · Click for next slide

The sigmoid maps any real-valued score into the interval (0, 1).

Probability and Class Labels

Step 1: estimate probability

The model returns p = σ(z).

Step 2: make a decision

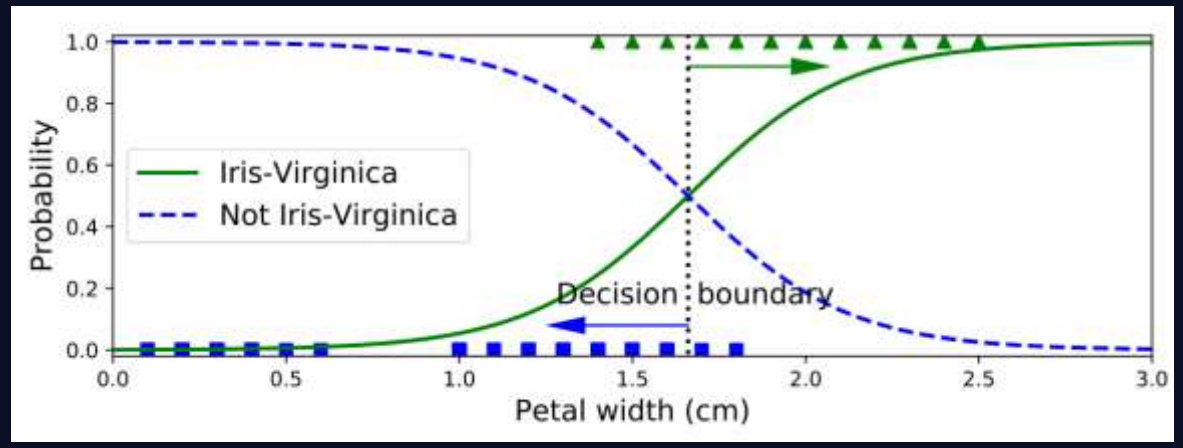
Compare p with a chosen threshold.

Rule used here	Predicted class
p > 0.5	1
p ≤ 0.5	0

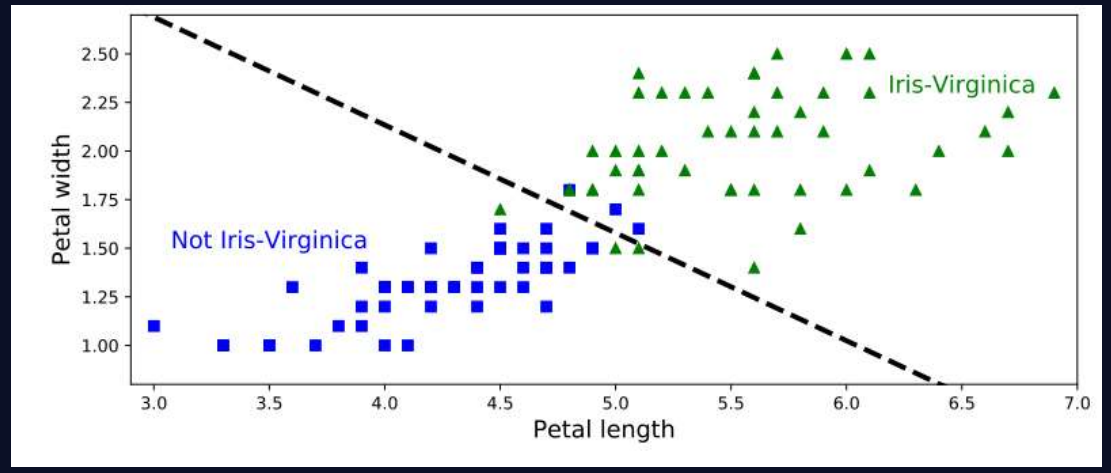
$$p = 0.5 \iff z = 0$$

The Decision Boundary

ONE FEATURE: A THRESHOLD



TWO FEATURES: A LINE



$$p = 0.5 \iff b + w_1 x_1 + w_2 x_2 = 0$$

Q19 • QUICK CHECK

With two original features and a linear score, the p = 0.5 boundary is…

A. Always an S-shaped curve

B. Always a circle

C. A straight line

D. Not defined

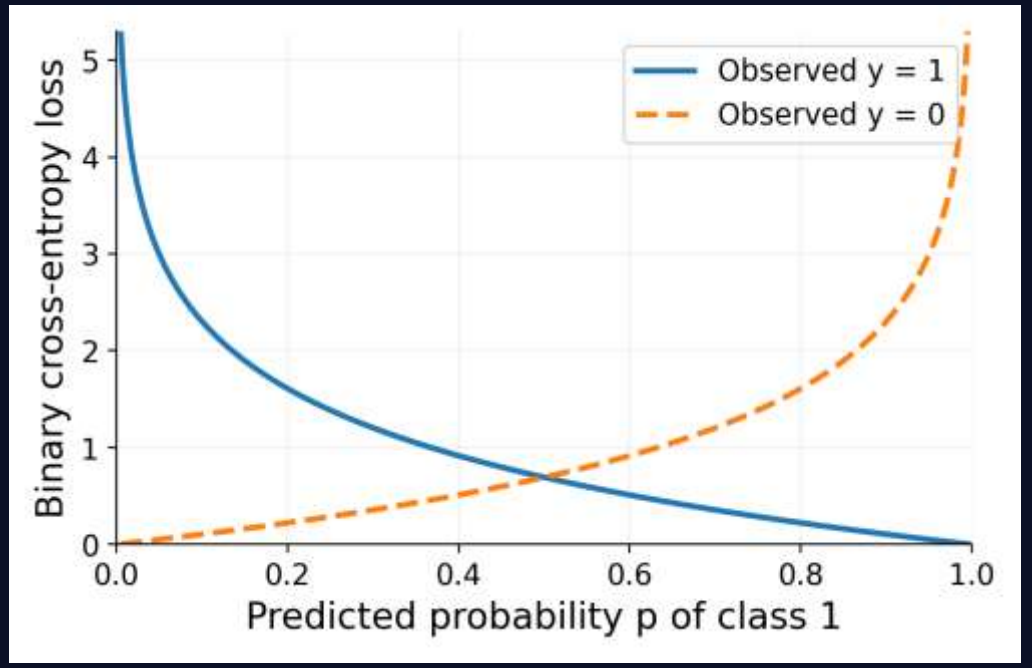
Binary Cross-Entropy

OBSERVED Y = 1

$$\ell = -\ln p$$

OBSERVED Y = 0

$$\ell = -\ln(1 - p)$$



$$\ell(y, p) = -[y \ln p + (1 - y)\ln(1 - p)]$$

Observed y = 1: Which gives less loss—p = 0.75 or p = 0.25?

Cross-Entropy Calculations

$$J_{\text{logistic}} = \frac{1}{N}\sum_{i=1}^{N} \ell(y_i, p_i)$$

y	p	Probability of observed class	Loss
1	0.75	0.75	0.288
1	0.25	0.25	1.386
0	0.75	1 − 0.75 = 0.25	1.386
0	0.25	1 − 0.25 = 0.75	0.288

The meaning of −ln

ln(1) = 0.

Lower probability for the observed class means higher loss.

Q18 • QUICK CHECK

For observed y = 1, which p gives the largest cross-entropy loss?

A. 0.01

B. 0.10

C. 0.50

D. 0.99

Training Logistic Regression

Predict

$z_i \rightarrow p_i$

Measure

Cross-entropy

Differentiate

Use $p_i - y_i$

Update

Subtract α × gradient

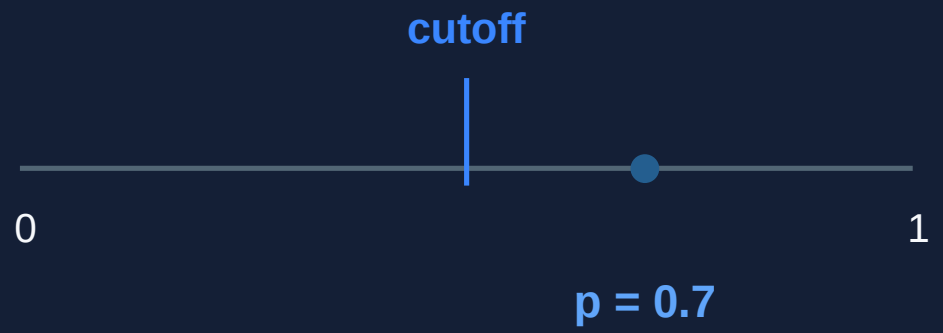
$$\frac{\partial J}{\partial w_j} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i) x_{ij}$$

$$\frac{\partial J}{\partial b} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)$$

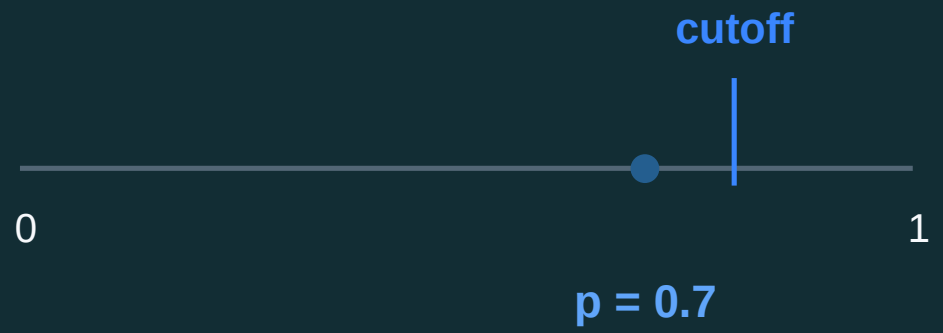
Train with probabilities. Apply a decision threshold when predicting labels.

Changing the Decision Threshold

Threshold τ = 0.5



Threshold τ = 0.8



Q17 • QUICK CHECK

p = 0.7; threshold changes from 0.5 to 0.8. With fixed parameters…

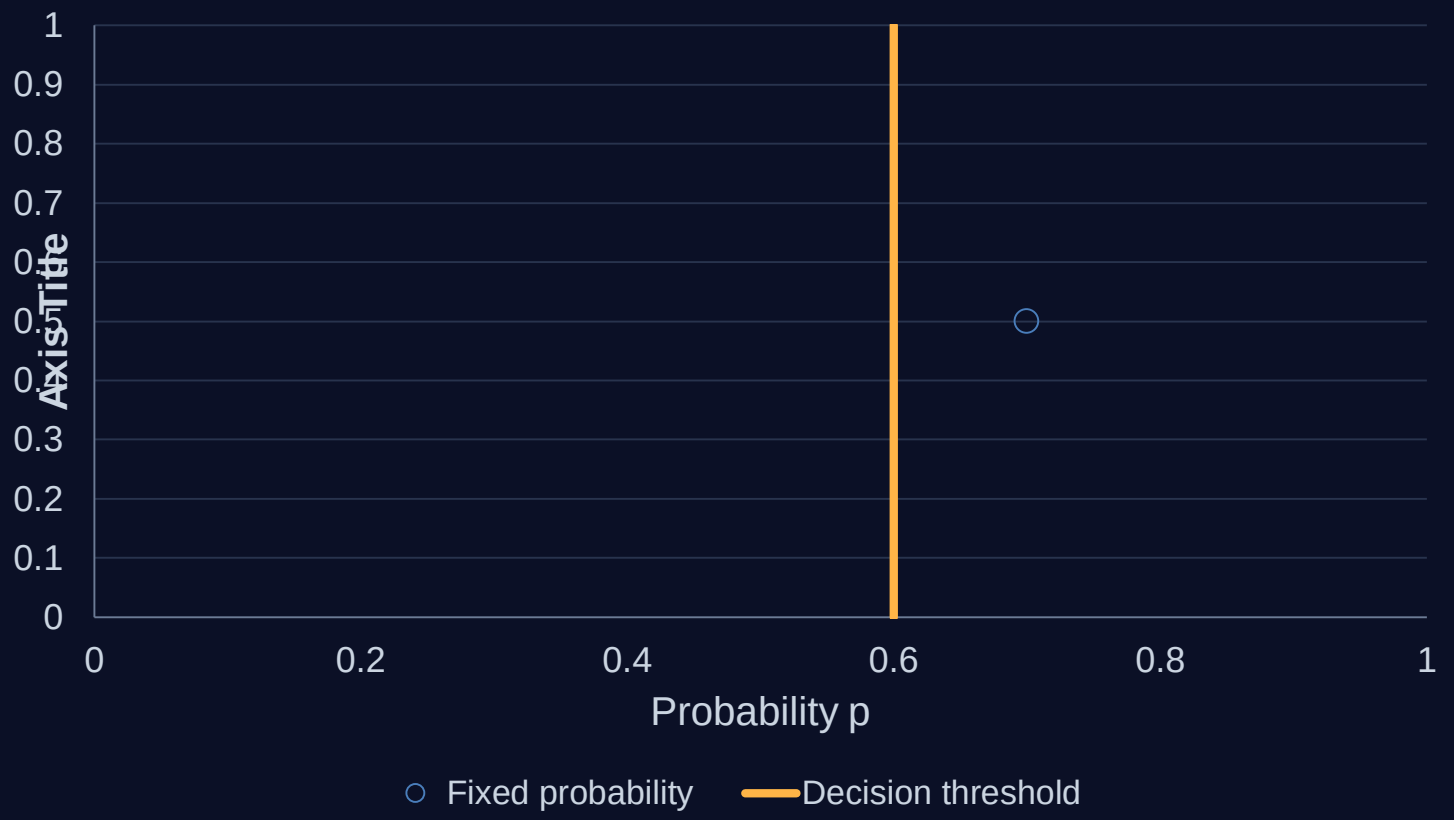
A. The probability becomes 0.8

B. Nothing changes

C. The model must be retrained first

D. p stays 0.7, but the predicted class changes

Animation: Changing the Decision Threshold



The model stays fixed

p = 0.70
Threshold τ = 0.60

Predicted class = 1

Rule: class 1 if p > τ.
Otherwise, class 0.

Step 3 of 3 · Click for next slide

With fixed model parameters, changing the threshold changes the decision.

Linear and Logistic Regression

Component	Linear regression	Binary logistic regression
Target	Numerical quantity	Class label
Prediction	$\hat{y} = b + w^T x$	$p = \sigma(b + w^T x)$
Objective used here	Half-MSE	Average cross-entropy
Gradient descent?	Yes	Yes
Label threshold?	Not needed	Needed for a hard label

Q20 • QUICK CHECK

Which objective pairing is used in this lecture?

A. Linear: cross-entropy; logistic: squared error

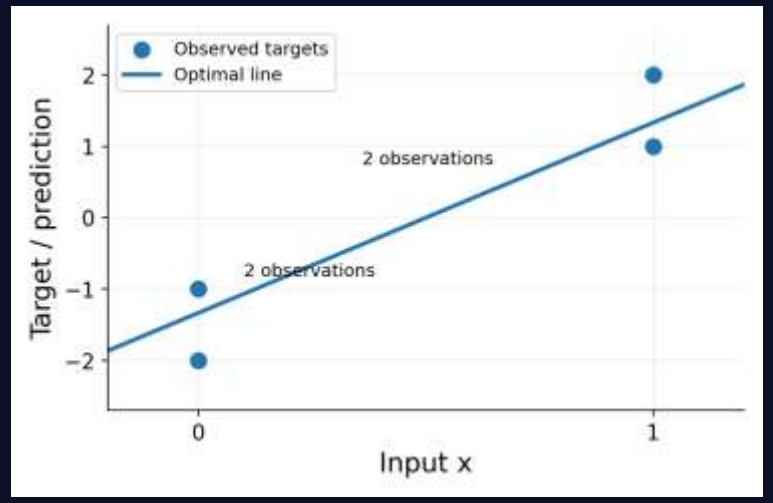
B. Linear: squared error; logistic: cross-entropy

C. Both must use 0–1 loss

D. Neither model needs an objective

Reading Three Different Graphs

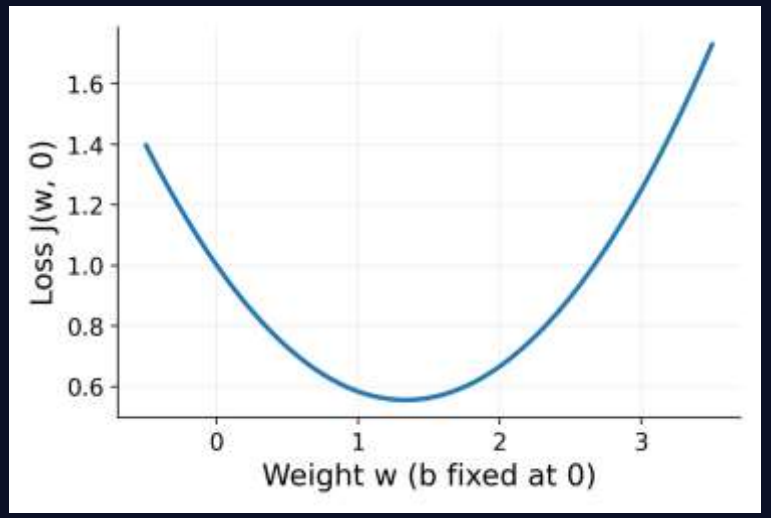
PREDICTION PLOT



Horizontal: input x

Vertical: target / prediction

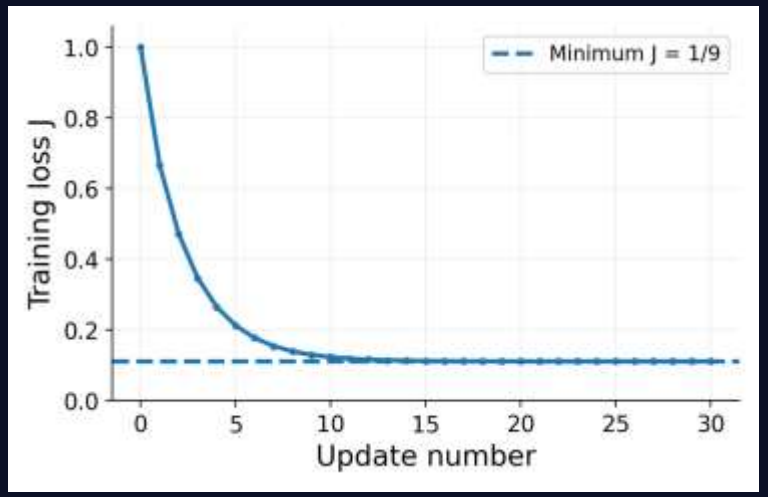
LOSS LANDSCAPE



Horizontal: parameter w

Vertical: J(w, 0); b fixed

TRAINING HISTORY



Horizontal: update number

Vertical: current training loss

Which graph answers “What do we predict?” and which answers “Are we learning?”

The Learning Pipeline

Group A: predict an amount

Advertising spending → sales

Group B: predict a class

Petal width → Virginica or not

1. **Name the input and the target.**
2. Write the model prediction.
3. Choose the training objective.
4. Explain one parameter update.
5. Is a decision threshold needed?

Model → objective → parameter updates → evaluation on unseen data

Instructor Appendix

Answers and worked solutions

55–58: all 20 MCQs

59: compare candidate lines

60: second gradient-descent update

61: probability and loss checks

Mathematics and provenance

62: least-squares solution

63: cross-entropy gradient

64: readings and figure credits

Speaker notes: teaching cues and full explanations

Logistic training follows Murphy’s cross-entropy—not Berkeley’s squared-error variant.

MCQ Answer Key: Q01–Q05

Q01	B — The target values are category codes, not numerical amounts.
Q02	C — Supply new features; use parameters learned during training.
Q03	A — Adding to b shifts every prediction upward without changing the slope.
Q04	D — Four feature weights plus one intercept gives five parameters.
Q05	C — $((+1)^2 + (-1)^2) / 2 = 1$. Squared errors do not cancel.

MCQ Answer Key: Q06–Q10

Q06	B — MSE = SSE / N = 2 / 6 = 1/3. Half-MSE J = 1/6.
Q07	C — A global minimum compares every candidate in the allowed domain.
Q08	A — The axis is a parameter, not an input feature or observation index.
Q09	D — Convexity rules out worse local minima; uniqueness needs full column rank.
Q10	B — Subtracting a positive multiple of a positive derivative decreases w.

MCQ Answer Key: Q11–Q15

Q11	A — $w_{new} = 0 - 1(-2/3) = 2/3$.
Q12	D — Compute both gradients at the same old parameter values, then update.
Q13	C — An oversized step can oscillate or diverge, even on a convex objective.
Q14	D — Fit on training data, select on validation, and reserve the test set.
Q15	B — The sigmoid gives the estimated probability of class 1.

MCQ Answer Key: Q16–Q20

Q16	C — $e^0 = 1$, so $\sigma(0) = 1/(1+1) = 0.5$.
Q17	D — p remains 0.7. The label changes from 1 at τ=0.5 to 0 at τ=0.8.
Q18	A — For y=1, loss = $-\ln(p)$, so p=0.01 has the greatest penalty.
Q19	C — The boundary is $b + w_1x_1 + w_2x_2 = 0$, a line in two original features.
Q20	B — Linear regression: half-MSE. Logistic regression: binary cross-entropy.

Activity Key: Candidate Lines

$$A:\ \hat{y} = 0 \qquad B:\ \hat{y} = 2x - 1$$

i	x	y	A: ŷ	A: error²	B: ŷ	B: error²
1	0	−1	0	1	−1	0
2	0	−1	0	1	−1	0
3	0	−2	0	4	−1	1
4	1	1	0	1	1	0
5	1	1	0	1	1	0
6	1	2	0	4	1	1

$$A:\ \mathrm{SSE} = 12,\quad \mathrm{MSE} = 2,\quad J = 1 \qquad B:\ \mathrm{SSE} = 2,\quad \mathrm{MSE} = 1/3,\quad J = 1/6$$

Activity Key: The Second Update

$$w = \frac{2}{3}, \quad b = 0, \quad \alpha = 1$$

i	x	y	ŷ	e	ex
1	0	−1	0	1	0
2	0	−1	0	1	0
3	0	−2	0	2	0
4	1	1	2/3	−1/3	−1/3
5	1	1	2/3	−1/3	−1/3
6	1	2	2/3	−4/3	−4/3

$$g_w = \frac{-2}{6} = -\frac{1}{3}, \qquad g_b = \frac{2}{6} = \frac{1}{3}$$

$$w_{\text{new}} = \frac{2}{3} - \left(-\frac{1}{3}\right) = 1$$

$$b_{\text{new}} = 0 - \frac{1}{3} = -\frac{1}{3}$$

$$\text{SSE}_{\text{new}} = \frac{17}{3}, \qquad J_{\text{new}} = \frac{17/3}{12} = \frac{17}{36} \approx 0.472$$

Activity Key: Probabilities and Losses

z	σ(z)	Label at threshold 0.5
−2	0.119	0
0	0.500	0 (tie rule)
2	0.881	1

Keep p = 0.7 fixed

Threshold 0.5 → label 1

Threshold 0.8 → label 0

WHEN THE OBSERVED TARGET IS Y = 1

$$-\ln(0.75) \approx 0.288 \quad < \quad -\ln(0.25) \approx 1.386$$

The Least-Squares Solution

TWO INPUT GROUPS

$$x = 0: \quad b = \frac{-1-1-2}{3} = -\frac{4}{3}$$

$$x = 1: \quad w + b = \frac{1+1+2}{3} = \frac{4}{3}$$

$$w = \frac{8}{3}, \quad b = -\frac{4}{3}, \quad J_{\min} = \frac{1}{9}$$

GENERAL NORMAL-EQUATION FORM

$$\hat{\theta} = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$$

Requires full column rank.

Here X has columns x and 1;
θ contains w and b.

The Cross-Entropy Gradient

$$\frac{d}{dp}\ln p = \frac{1}{p}, \qquad p = \sigma(z), \qquad \frac{dp}{dz} = p(1-p)$$

1 DIFFERENTIATE THE LOSS WITH RESPECT TO P

$$\frac{\partial \ell}{\partial p} = -\frac{y}{p} + \frac{1-y}{1-p}$$

2 MULTIPLY BY THE SIGMOID DERIVATIVE

$$\frac{\partial \ell}{\partial z} = \left[-\frac{y}{p} + \frac{1-y}{1-p}\right] p(1-p) = -y(1-p) + (1-y)p = p - y$$

3 MULTIPLY BY THE LINEAR-SCORE DERIVATIVE

$$\frac{\partial \ell}{\partial w_j} = (p-y)x_j, \qquad \frac{\partial \ell}{\partial b} = p - y$$

Readings and Visual Credits

Kevin P. Murphy

Probabilistic Machine Learning: An Introduction — Chapters 10 and 11

Support: §§1.2, 2.4, 4.5, 7.8, 8.1–8.2, 8.4

Harvard CS50 AI

Lecture 4: Learning

cs50.harvard.edu/ai/notes/4/

UC Berkeley CS188

Textbook §§1.5 and 9.5–9.7

Official online textbook topic sections

Visuals: attributed textbook panels + graphs calculated from stated equations.

Classroom adaptations, chosen coefficients, and probe values are labeled throughout.